



ONLYIAS
Nothing Else

IDMP TEST SERIES - 2022

High Yielding PRELIMS TEST SERIES

Test – 20

Environment & Ecology

Topics covered:

- BIODIVERSITY, INDIAN BIODIVERSITY, SCHEDULE ANIMALS OF WPA, 1972
- PROTECTED AREA NETWORK, CONSERVATION EFFORTS
- CLIMATE CHANGE, MITIGATION STRATEGIES
- OZONE DEPLETION, OCEAN ACIDIFICATION, INDIA AND CLIMATE CHANGE
- CLIMATE CHANGE ORGANISATIONS
- Current Affairs August 2021



Scan the QR code for Schedule

DEDICATED PRELIM PROGRAM(DPP)**UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022****DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO****TEST BOOKLET SERIES****Test Code:****DPP****TEST BOOKLET
GENERAL STUDIES
TEST - 20****E 20****Time Allowed: Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

TOPICS COVERD: BIODIVERSITY, INDIAN BIODIVERSITY SCHEDULE ANIMALS OF WPA 1972, PROTECTED AREA NETWORK, CONSERVATION EFFORTS, CLIMATE CHANGE, MITIGATION STRATEGIES, OZONE DEPLETION, OCEAN ACIDIFICATION, INDIA AND CLIMATE CHANGE, CLIMATE CHANGE ORGANISATIONS, Current Affairs August 2021



Q.1) Sonai-Rupai Wildlife Sanctuary, Orang National Park, Barnadi Wildlife Sanctuary and recently included Manas National National Park are known for the conservation and reintroduction of which of the following species?

- a) Spider monkey
- b) Indian Giant Squirrel
- c) Pygmy Hog
- d) Marbled Cat

Q.2) Consider the following statements with respect to Kaziranga National Park:

- 1. In March 2020, it was selected as one among the 17 Iconic Tourist Sites of the country by the Indian Government.
- 2. Presence of alluvial savanna woodlands and tropical semi-evergreen forest vegetation in the National Park.
- 3. It is designated as an 'Important Bird Area' by Birdlife International.
- 4. It is located in Chirang and Baksa District of Assam in northeast India and is contiguous with the National Park of Bhutan.

Which of the above statements is/are correct ?

- a) 1, 2 and 3 only
- b) 2 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

Q.3) With respect to community reserves in India, consider the following statement:

- 1. To be notified as a Community Reserve, the land should be community land or land owned by the State Government.
- 2. There is no provision for change in land use pattern within a Community Reserve at any time.
- 3. Nagaland has the maximum number of community reserves in India in terms of area.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 and 3 only
- c) 1, 2 and 3
- d) 3 only

Q.4) Management Effectiveness Evaluation (MEE) of National Park and Wildlife Sanctuaries in India is released by which of the following organizations?

- a) Ministry of Environment, Forest and Climate Change

- b) NITI Aayog.
- c) The Energy and Resource Institute.
- d) WildLife Trust of India.

Q.5) Soaring wildlife smuggling is a big threat to their conservation. Which of the following entities enforces the anti-smuggling laws in India?

- a) Directorate of Revenue Intelligence.
- b) WWF-India.
- c) Wildlife Crime Control Bureau (WCCB).
- d) TRAFFIC-India

Q. 6) Consider the following statements regarding 'Conservation Reserve Management Committee':

- 1. It is constituted under the Forest Conservation Act, 1980.
- 2. The State Government shall constitute a conservation reserve management committee to advise the Chief Wildlife Warden.
- 3. Change in the land use pattern can be made within the community reserve without the permission of the Conservation Reserve Management Committee.

Select the correct statement(s) from the codes given below.

- a) 1 and 2 only
- b) 1 only
- c) 2 only
- d) 2 and 3 only

Q.7) Consider the following marine protected areas of India:

- 1. Gulf of Kutch Marine National Park
- 2. Mahatma Gandhi Marine National Park
- 3. Gulf of Mannar Marine National Park
- 4. Rani Jhansi Marine National Park

Which is their correct order from north to south?

- a) 1-2-4-3
- b) 1-4-2-3
- c) 1-2-3-4
- d) 1-4-3-2

Q.8) It is an artificial lake, and has recently been designated as a Ramsar site. Originally created to supply water to the city of Agra in the summer, the wetland soon became an important and rich



ecosystem. Alongside this lake lies a sloth bear rescue facility dedicated to rehabilitating previously enslaved 'dancing bears'. The Site is important for bird species which migrate on the Central Asian flyway, with over 30,000 waterbirds known to visit the reservoir annually.

Which of the following sites best represent the above description?

- a) Nawabganj Bird Sanctuary
- b) Samaspur Bird Sanctuary
- c) Parvati Aranga Bird Sanctuary
- d) Sur Sarovar Bird Sanctuary

Q.9) With reference to the sacred groves in India, consider the following pairs:

- 1. Sarana - Rajasthan
- 2. Jogmaya - Jharkhand
- 3. Gumpa Forests - Arunachal Pradesh
- 4. Pavithravana - Andhra Pradesh

Which of the pairs given above are correctly matched ?

- a) 1 and 2
- b) 2 and 3
- c) 2 and 4
- d) 3 and 4

Q.10) Consider the following National Parks:

- 1. Bandipur
- 2. Nagarhole
- 3. Silent Valley
- 4. Banerghatta

Arrange the above from low latitude to high latitude.

- a) 4-2-1-3
- b) 2-4-1-3
- c) 3-1-4-2
- d) 3-1-2-4

Q.11) Which of the following statements best describes the 'deemed forests'?

- a) Thickly wooded areas of the revenue department
- b) Land having characteristic of forests irrespective of ownership
- c) Thickly/sparsely wooded areas of the forest department
- d) All recorded land of forest department irrespective of vegetation status

Q. 12) 'SAVE' is a conservation initiative to protect which of the following species?

- a) Snow Leopard
- b) Great Indian Bustard (GIB)
- c) Vultures
- d) Sea Turtle

Q.13) Consider the following statements about the 'No-Go' forests:

- 1. No mining activities are allowed in such forests.
- 2. They are designated so under the Wildlife Protection Act, 1972.
- 3. These areas have more than 20 percent weighted forest cover (WFC) or more than 30 per cent gross forest cover (GFC).

Select the correct answer using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.14) 4th cycle of All India Tiger Estimation was released sometime back. In this context, consider the following statements:

- 1. Pench Tiger Reserve in Madhya Pradesh recorded the highest number of tigers.
- 2. Sathyamangalam Tiger Reserve in Tamil Nadu registered the "maximum improvement" since 2014.
- 3. No tigers found in Buxa, Valmiki and Dampa reserves.
- 4. India is home to nearly 75% of the global tiger population.



Select the correct answer using the codes given below.

- a) 1, 2 and 4
- b) 1, 3 and 4
- c) 2, 3 and 4
- d) All of the above

Q.15) It is situated at the confluence of Banas and Chambal in the vicinity of the Aravali hills and Vindhya plateau. It is also a Tiger Reserve. Jogi Mahal is another attraction of this place. Many picturesque lakes enrich the landscape.

Identify the protected area.

- a) Madhav National Park
- b) Gandhi Sagar Wildlife Sanctuary
- c) Ram Sagar Wildlife Sanctuary
- d) Ranthambore National Park

Q.16) Which among the following protected area sites has the unique distinction of being a national park, a wildlife sanctuary, a world heritage site, a biosphere reserve, a tiger reserve as well as an elephant reserve?

- a) Kaziranga National Park
- b) Panna National Park
- c) Bandipur National Park
- d) Manas National Park

Q.17) "India Eco Development Project" is related to the development of which of the following ?

- a) For the development of the National Waterways of India.
- b) To conserve biological diversity in globally significant protected areas.
- c) For the protection of Eastern Himalayan Ecosystem.
- d) To counter the frequent landslides in the Himalayan States by checking on unplanned development of the hilly regions.

Q.18) Global Tiger Initiative is a program initiated by which of the following organizations ?

- a) International Conservation of Nature

- b) World Wide Fund for Nature
- c) World Bank
- d) Wildlife Institute of India.

Q.19) Consider the following statements about the Graded Response Action Plan:

- 1. Graded Response Action Plan works only as an emergency measure.
- 2. The plan is incremental in nature.
- 3. The GRAP is to be implemented across all the states of India including Jammu and Kashmir.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) 2 only

Q.20) In the context of Forest Conservation, consider the following statements:

- 1. Net Present Value is a voluntary quarterly payment that a user has to make for diverting forestland for non-forest use.
- 2. These payments go to the Compensatory Afforestation Fund, managed by the Ministry of Environment, Forest and Climate Change.
- 3. The Gram Sabha decides on whether forests can be diverted for projects and the NPV to be charged.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) None of the above

Q.21) With reference to Renewable Energy, consider the following statements:

- 1. The renewable energy capacity in India is currently 1000 GigaWatts.
- 2. The UJALA scheme aims to promote solar power in the western part of the country.
- 3. Smart grids have self-healing capabilities and enable electricity customers to become active participants.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above



Q.22) Consider the following statements about the Adaptation Fund:

1. It was established under the Kyoto Protocol of UNFCCC.
2. The Clean Development Mechanism is a source of income for the Adaptation Fund.
3. The World Economic Forum serves as trustee of the Adaptation Fund on an interim basis.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) All of the above

Q.23) Consider the following statements:

1. Gasification is a process that can convert any carbon-based raw material such as coal into fuel gas.
2. Syngas is a mixture of hydrogen (H₂), carbon monoxide (CO) and carbon dioxide (CO₂).

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.24) In the context of Flue Gas Desulphurization, consider the following statements:

1. It is a set of technologies used to remove sulfur dioxide and carbon dioxide from exhaust flue gases of fossil-fuel power plants.
2. Lime is used to remove acidic gases, particularly sulfur dioxide from flue gases.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.25) With reference to Climate Smart Agriculture (CSA), consider the following statements:

1. Permaculture and Monoculture are essential components of Climate Smart Agriculture.
2. Conservation tillage systems cause significantly less soil erosion due to wind and water.

3. Improving farmer productivity, and as a result, livelihoods is one of the goals of CSA.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only
- d) None of the above.

Q.26) In the context of Stockholm Convention Persistent Organic Pollutants, consider the following statements:

1. India is yet to ratify the Stockholm Convention.
2. The Global Environmental Facility is designated as the financial mechanism for the Stockholm Convention.
3. The convention calls to limit the use of DDT to malaria control.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1 and 3 only

Q.27) In the context of Green Crackers, consider the following statements:

1. Green crackers are produced using no harmful raw materials and have additives which eliminate emissions completely.
2. They contain chemicals such as lithium, arsenic, barium and lead.
3. Green crackers release water vapour and don't allow the dust particles to rise.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.28) With reference to Solar Radiation Management, consider the following statements:

1. Its aim is to reduce the inflow of solar energy, rather than to reduce global warming by reducing greenhouse gases.
2. Stratospheric aerosol injection and Whitening clouds could reflect away inbound sunlight and cool the planet.



3. Low and thick clouds reflect shortwave radiation back to space.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.29) Consider the following statements about enhancing Climate Change Mitigation through Agriculture:

- 1. Integrating livestock into crop production systems reduces agriculture's own footprint.
- 2. Increased tillage can contribute to cutting the greenhouse-gas emissions of agriculture.
- 3. Decreasing meat production can help in climate change mitigation.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) All of the above

Q.30) Consider the following statements:

- 1. Pluviculture is the act of artificially increasing precipitation/rain.
- 2. It is a permanent way of suppressing the pollutants present in the air.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.31) Consider the following statements:

- 1. Only Developed nations are required to reduce HFC use as per the Kigali Amendment to the Montreal Protocol.
- 2. Hydro Fluoro Carbons are more potent than Carbon Dioxide.
- 3. Unlike the Paris Agreement, the Kigali Amendment to the Montreal Protocol is legally binding with mandatory HFC reduction targets for countries.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only

- d) None of the above

Q.32) Consider the following statements:

- 1. The government of Bhutan is the first country to become carbon negative.
- 2. Carbon Neutrality involves zero emission of Carbons by the industries.
- 3. Greenwashing can help in achieving Carbon Neutrality.

Which of the statements given above is/are correct?

- a) 2 only
- b) 2 and 3 only
- c) 1 only
- d) All of the above

Q.33) Consider the following statements about Paris Climate Accord:

- 1. It is an agreement within the United Nations Framework Convention on Climate Change.
- 2. Like the Kyoto Protocol, the Paris Agreement, allows for voluntary and nationally determined targets.
- 3. Iran, Iraq and Libya, China and South Sudan have not ratified the Paris Agreement.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.34) Consider the following Pairs:

Treaties/Protocols	Components
1. REDD+	- Focuses on eliminating transboundary movement of exotic animals.
2. Nagoya Protocol	- Applies to Human Genetic Material.
3. Talanoa Dialogue	- Guidance for proper importing of living modified organisms.

Which of the pairs given above is/are correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) None of the above.

Q.35) With reference to the State Pollution Control Boards (SPCB), consider the following statements:



1. The SPCB is constituted under the Water Act, 1974.
2. SPCB has the power to hold, acquire, and dispose of property.
3. SPCB has to inspect sewage or the trade effluents works and plants to treat sewage and trade effluents.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.36) In the context of Fuel Cell Electric Vehicles (FCEV), consider the following statements:

1. FCEVs rely on a constant supply of hydrogen and oxygen.
2. FCEVs do not require charging of the vehicles.
3. FCEVs emit only water vapour and warm air.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) All of the above

Q.37) National Innovation on Climate Resilient Agriculture is launched by which of the following organizations?

- a) Food and Agriculture organization (FAO)
- b) Indian Council of Agricultural Research (ICAR)
- c) Ministry of Environment Forest and Climate Change (MoEFCC) in association with Niti Aayog.
- d) National Bank for Agriculture and Rural Development.

Q.38) Recently in news the Pensilungpa Glacier, is located at

- a) Eastern Karakoram Range
- b) Kumaon Himalayas
- c) Lahaul and Spiti
- d) Zaskar region of Ladakh

Q.39) Which of the following schemes for upliftment of the minorities works under the aegis of the Union Ministry of Minority Affairs ?

1. Nai Manzil
2. Nai Roshni
3. Naya Savera
4. Padho Pardesh
5. Hamari Dharohar

6. USTTAD (Upgrading the Skills and Training in Traditional Arts/Crafts for Development).

Select the correct answer from the codes given below:

- a) 1, 2, 4 and 5 only
- b) 2, 3, 5 and 6 only
- c) 1, 4 and 5 only
- d) 1, 2, 3, 4, 5 and 6

Q.40) With reference to the recently released Global Youth Tobacco Survey (GYTS-4) India 2019, consider the following statements:

1. There is decline in decline in tobacco use school going children (13- 15 years) in last decade
2. The survey was conducted under the administrative care of the Union Ministry of Women and Child Development.
3. Tobacco use among school going children (13-15 years) is highest in North Indian States of India.

Which of the above statements is/are correct ?

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.41) Which of the following statements is incorrect with reference to 'Jan Aushadi Sugam App' ?

- a) The government launched the Jan Aushadi Sugam App to ensure the availability of quality generic medicines at affordable prices.
- b) The Union Ministry of Health and Family Welfare is the nodal ministry that developed and tracks the usability of this application.
- c) The application also locates the nearest Pradhan Mantri Bharatiya Janaushadhi Kendra where these medicines can be purchased.
- d) None of the above statements is incorrect.

Q.42) Madhav National Park- Madhya Pradesh, Sitamata Wildlife Sanctuary- Rajasthan, Mukundra Hills Tiger Reserve- Rajasthan, Gandhi Sagar Wildlife Sanctuary- Madhya Pradesh recently in news due to

- a) Asiatic Lion Reintroduction Project
- b) Great Indian Bustard Rehabilitation Programme.
- c) Cheetah Reintroduction Programme.
- d) Important Bird Area designated by Bird Life International.

Q.43) With reference to 'Biodiversity Hotspot' consider the following statements

1. Biodiversity hotspots are a Biogeographic region with a significant amount of reservoir of



Biodiversity which are not yet threatened with Destruction

2. Conservation International sets criteria for determination of a Biodiversity hotspot
3. Among other criterias, Only Plant population and not animal population is considered in the criteria for determination of a Biodiversity hotspot
4. India shares four major biodiversity hotspots

Which of the above statements is/are correct ?

- a) 1, 2 & 3 only
- b) 2, 3 & 4 only
- c) 1, 2 & 4 only
- d) 1 & 4 only

Q.44) Consider the following statements :

1. Species evenness represents the numerical equality of species
2. Species richness is measured for a multiple species in a community
3. Low species evenness with high species richness indicated towards a community where a few species dominates

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 & 3 only
- c) 2 only
- d) 1, 2 & 3 only

Q.45) Consider the following statements regarding 'Invasive Species' and choose the Incorrect Statement

- a) An Invasive species can be any kind of living organism that is not native to an ecosystem. And causes harm
- b) India's National Biodiversity Target 4 is to identify Invasive alien species
- c) Not all Non-Native species are Invasive
- d) Ministry of Environment, Forests and climate change issues approvals for the export of wild animals under Biological Diversity Act 2002

Q.46) Match the following statements with suitable options

1	Alpha Diversity	(i)	It is the species diversity between two adjacent ecosystems and is obtained by comparing the number of
---	-----------------	-----	--

			species unique to each of the ecosystems.
2	Gamma Diversity	(ii)	It is measured by counting the number of taxa (usually species) within a particular area, community or ecosystem.
3	Beta Diversity	(iii)	It refers to the diversity of the habitats over the total landscape or geographical area.

Choose the Correct option

- a) 1-i ; 2- ii; 3-iii
- b) 1- ii ; 2 - i; 3 - iii
- c) 1- ii, 2- iii; 3- (i)
- d) 1 - iii, 2 - i, 3 - (ii)

Q.47) Consider the following statements

1. Inbreeding within a population can decrease individual variability
2. Low genetic variation depresses resistance to diseases
3. Genetic diversity occurs only between species and not within species

Which of the above statements is/are correct

- a) 1 & 2 only
- b) 3 only
- c) 1 only
- d) 1, 2 & 3

Q.48) With reference to species diversity, consider the following statements

1. Species diversity of zero means there is only one species present
2. Species diversity usually decreases as we move away from the equator towards the poles
3. Temperate regions, as an exception, harbour more species due to its climate, than polar, tropical and equatorial region

Which of the following statements is/are correct?

- a) 1 & 3 only
- b) 2 only
- c) 3 only
- d) None of the above

Q.49) The species selected to act as an ambassador, icon or symbol for a defined habitat, issue, campaign or environmental cause and by focussing on them,



their habitat and other species which share the same habitat can also be conserved

Above statement represents which of the following ?

- a) Umbrella Species
- b) Iconic Species
- c) Flagship Species
- d) Guilds

Q.50) Consider the following statements

- 1. Keystone species perform a check and balance over the population of other species in an ecosystem.
- 2. Keystone Species are also called Ecosystem Engineers
- 3. Keystone Species have high functional redundancy

Which of the following statement/s is/are correct

- a) 1 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) None of the above

Q.51) Consider the following statements

- 1. Foundation species are Dominant Consumer in an ecosystem
- 2. The term foundation species is used for a species with a strong role of restructuring a community
- 3. Foundation species can occupy any trophic level

Which of the above statement is/are correct?

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q.52) The Indian region is composed of two realms.

Which of the following realms are not part of Indian region ?

- 1. Palaearctic Realm
- 2. Neotropical Realm
- 3. Indomalayan Realm
- 4. Australian Realm

Select the correct answer using the codes given below.

- a) 1 & 2
- b) 2 & 4
- c) 1 & 3
- d) 3 & 4

Q.53) Consider the following statements with reference to Bio-geographic zone

- 1. Of the ten Biogeographic regions of India, Gangetic plain account for second largest in terms of % of total geographical area after Deccan plateau
- 2. Andaman and Nicobar Island shows high Endemism in terms of flora and fauna
- 3. Wildlife Institute of India (WII) outlined a scheme to divide India into different Biogeographic zones in 1986

Choose the correct option

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q.54) Consider the following statement

- 1. Acclimatization is the process in which an individual organism adjusts to changes in its environment
- 2. Acclimatization takes a long period of time and takes generations to make genetic adjustments

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2

Q.55) Human activity along with Natural cycles over the past few centuries has led to the extinction of several hundred species. In that Reference, consider the following species

- 1. Steller's Sea Cow
- 2. Quagga
- 3. Caspian tiger
- 4. Sloth bear
- 5. Thylacine
- 6. Bali Tiger
- 7. Himalayan Quail

Which of the above species are extinct ?

- a) 1, 2, 3, 4, 5 & 7 only
- b) 1, 2, 4 & 6 & 7 only
- c) 1, 2, 3, 5 & 6 & 7 only
- d) 1, 2, 3, 4 & 6 only

Q.56) Consider the following statements

- 1. Biodiversity Hope spots are the world's last refuges where high numbers of threatened species still persist



2. Biodiversity Cool spots is an area of an ocean that needs special protection because of its wildlife and significant underwater habitats
3. India doesn't host any hope spots or cool spots

Choose the correct Statement

- a) 1 only
- b) 1 & 2 only
- c) 1 & 3 only
- d) None of the above

Q.57) Consider the following statements with reference to Protected Areas under Wildlife Protection Act 1972

1. Both Central as well as state govt can declare an area as National Park or Wildlife Sanctuary
2. Legislation must be passed with a simple majority to alter boundary of a National park or wildlife sanctuary
3. National Board for wildlife make recommendations on setting up & management of protected area

Which of the following statements is/are true ?

- a) 1 & 2 only
- b) 1 & 3 only
- c) 2 & 3 only
- d) 1, 2 & 3

Q.58) With reference to National park and wildlife sanctuary , consider the following statements

1. Grazing activity is not permitted in National Park
2. Wildlife sanctuary is formed for particular Species whereas National park may or may not be for particular Species
3. National Park can be downgraded to wildlife sanctuary

Which of the following statements is/are correct ?

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q.59) With reference to community reserve, Consider the following statements

1. It is declared by the central government in consultation with the state government
2. It is declared in a privately owned land which is not within a National park, sanctuary or conservation reserve
3. Community Rights of people, such as grazing etc, in such reserve are not affected

Choose the correct statement

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q. 60) With reference to Conservation reserve, Consider the following statement :

1. It is declared by the State Governments in any area owned by central or State Government
2. There are limitations on rights of people in these reserves
3. Tamilnadu was the first state to get a conservation reserve
4. Developmental activities like roads or recreational activities can't be taken in Conservation Reserve

Which of the following statements is/are correct

- a) 1, 2 & 4 only
- b) 1 & 3 only
- c) 2 & 4 only
- d) 1 & 4 only

Q.61) With reference to Tiger Reserve, consider the following statements

1. NTCA (National Tiger conservation Authority) declares an area as Tiger reserve
2. Boundaries of Tiger Reserve can be altered with recommendations of NTCA
3. NTCA is a statutory body under the enabling provision of wildlife protection act 1972
4. St Petersburg Declaration was adopted in 2010 on Tiger conservation

Which of the following statements is/are correct

- a) 1, 2 & 3 only
- b) 1, 3 & 4 only
- c) 2, 3 & 4 only
- d) 1, 2, 3 & 4

Q.62) Consider the following statements with reference to Biosphere reserve

1. Biosphere reserves was declared as a part of UNESCO'S MAB (Man and Biosphere) Program in 1975
2. Buffer zone of Biosphere reserve is used for scientific research
3. Sand & stone mining are permitted in Biosphere reserve

Which of the following statements is/are correct

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only



d) 1, 2 & 3

Q.63) Consider the following statement

1. Sacred groves is a living example of religious / cultural tradition which emphasise on nature's protection
2. Botanical Garden is an ex-situ conservation whose one of the purpose is to study the introduction and acclimatisation process of exotic plants.
3. Zoo is a stationary or mobile establishment where captive animals are kept for exhibition

Which of the following statements is/are correct ?

- a) 1 only
- b) 2 & 3 only
- c) 3 only
- d) 1, 2 & 3

Q.64) With reference to wildlife protection act 1972, Consider the following statements

1. Wildlife protection act 1972 was passed by parliament under Article 252 of Indian constitution
2. None of the schedule under the act refer to plant species
3. Hunting of wild animal in schedule 1 can't be permitted in any case
4. Management of zoos is also under the ambit of wildlife Protection Act 1972

Choose the correct statement/s

- a) 1 & 2 only
- b) 1 & 3 only
- c) 2 & 4 only
- d) 1 & 4 only

Q.65) With reference to Rhinoceros, consider the following statements

1. All three species of Rhinoceros, Greater One horned Rhino, Sumatran Rhino and Javan Rhino are listed under Appendix I of CITES and Schedule I of Wildlife Protection Act 1972
2. Single Horn is found in Both male as well as female Indian Rhino(Greater One horned Rhinoceros)
3. India, Nepal, Bhutan, Malaysia and Indonesia are the five rhino range nations
4. Asia has five Species of Rhino - White Rhino, Black Rhino, Great One horned Rhino, Sumatran rhino and Javan Rhino

Choose the correct Statement using the codes give below

- a) 1 only

b) 2 only

c) Both 1 & 2

d) Neither 1 nor 2

Q.66) Which of the following statements is incorrect with respect to 'Dimethylsulphide'(DMS)?

- a) It is the most abundant biological sulphur compound emitted in the atmosphere mostly by the phytoplanktons.
- b) DMS encourages cloud formation.
- c) It is also known as the 'Anti GreenHouse Gas (GHG)'
- d) None of the above statements is incorrect.

Q.67) Consider the following statements with respect to Global Temperature:

1. Global temperature is an average of air temperature recordings from weather stations on land and sea as well as some satellite measurements
2. Globally the 2011- 2021 decade is considered as the warmest decade of the history of planet Earth.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.68) Recently in the news, HadCRUT4 is related to ?

- a) A group of Satellite systems used to track climate change and related aspects.
- b) An Early Warning System that gives information related to sudden the rise of 'Suspended Particulate Matters' in the atmosphere.
- c) The new technology of the filter system introduced for the industrial chimneys.
- d) Collection of combination of global land surface temperature data set.

Q.69) Which of the following statements correctly defines 'Global Warming Potential'?



- a) It is the ability of a greenhouse gas to absorb heat compared to carbon dioxide over a specified period of time.
- b) The potential of a greenhouse to trap heat from the sun.
- c) It refers to the retreat of sea ice and land ice mass from its historic extents.
- d) It is a measure of the reactivity of an individual Greenhouse Gas to the presence of other Greenhouse Gases that form a potentially stronger Greenhouse effect.

Q.70) The measure of distance useful for local winds, thunderstorms and tornadoes which ranges from a few to a few hundred miles to study climate change, is known as ?

- a) Planetary scale
- b) Mesoscale
- c) Synoptic scale
- d) Isohyet scale

Q.71) Consider the following statements with reference to 'Coupled Model Intercomparison Project (CMIP)':

- 1. It coordinates and archives climate model simulations based on shared model inputs by modelling groups from around the world.
- 2. The World Climate Research Programme (WCRP) is the parent organization for the development of the CMIP.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.72) Which one of the following statements correctly defines 'Mitigation Potential'?

- a) It is a measurement of the amount of carbon that can be stored in order to balance the release of carbon.
- b) It is the process to help predict how variables such as temperature, rain, and carbon dioxide levels get affected by the GHG.
- c) It is the process of addressing impacts to the environment caused by human action, notably

those resulting from highway, energy, water, and other infrastructure projects.

- d) It refers to efforts to reduce or prevent emission of greenhouse gases by using renewable energy.

Q.73) In the context of climate change, what does decoupling mean?

- a) Where economic growth and usage of fossil fuels grows but at different rates.
- b) Where economic growth happens but fossil fuels decline.
- c) Where economic growth is no longer strongly associated with consumption of fossil fuels.
- d) None of the above.

Q.74) With reference to the 'Climate Change Performance Index', consider the following statements:

- 1. It is published by The Climate Change Performance Index published by Germanwatch, CAN International and the NewClimate Institute
- 2. It is a biannual publication highlighting the tracking of the climate mitigation performance of countries across the globe
- 3. India ranks in the top 10 in terms of advances in climate change mitigation practices of the recently published index.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) 1 and 3 only.
- d) 1, 2 and 3

Q.75) The 'State of Global Climate' report is published by

- a) United Nation Framework Convention on Climate Change
- b) World Meteorological Organization
- c) United Nations Environment Programme
- d) World Bank.

Q.76) He was born in a village near Thiruvananthapuram, in 1856. His father affectionately called him Nanu. Since the childhood he had showed interest the great epic Ramayana and



Mahabharata. Later he wrote books named as Jathi Lakshanam, Advaita Deepika, Athma Vilasam
Which of the following person is described in the above passage?

- a) Erode Venkatappa Ramasamy
- b) Ramanujacharya
- c) Koyapalli Kelappan
- d) Sree Narayana Guru

Q.77) Changpa tribe recently sometimes in the news, found in which of the following states/Union territory ?

- a) Rajasthan
- b) Ladakh
- c) Arunachal Pradesh
- d) Lakshadweep

Q.78) consider the following statements about the Yuktadhara :

- 1. It is a geospatial planning portal which will help in facilitating new MGNREGA assets using remote sensing and geographic information system-based data.
- 2. It is launched by Ministry of Rural Development and Panchayati Raj.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.79) Ethnic group called 'Hazara' are in news sometimes, in Which of the following countries they are found ?

- a) Iran
- b) Iraq
- c) Afghanistan
- d) Saudi Arabia

Q.80) Which of the following statements is/are incorrect regarding the Prime Ministers Shram Awards ?

- 1. The objective of the Prime Ministers Shram Awards is to recognize the outstanding contributions made by workmen as defined in the Industrial Dispute Act, 1947 in organizations both in public and private sector.
- 2. Workmen solely engaged in routine service duties are not eligible for the awards

Select the correct answer using the codes given below.

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.81) Recently the Government launched SUJALAM', a '100 days campaign' as part of the 'Azadi Ka Amrit Mahotsav' celebrations, which of the following is objective of it ?

- a) To create more and more ODF Plus villages by undertaking waste water management at village level.
- b) To incentivize the States to increase investment in Agriculture and allied sectors to achieve 4% growth in agriculture sector.
- c) To ensure socio-economic development in the North-Eastern Region of India, by promoting tourism.
- d) None of the Above

Q.82) What is the significance of places called Thol, Bhindawas and Wadhwana, that were in the news recently ?

- a) Saline Lakes
- b) Glaciers
- c) Ramsar sites
- d) rivers in North East India

Q.83) With reference to Sovereign Gold Bond Scheme, consider the following statements:

- 1. It will drastically reduce the India's dependence on gold import.
- 2. Holder of the bond will earn some fixed interest, payable semi annually, which is tax free.
- 3. Sovereign gold bond can be used as collateral to get loan.



Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 3 Only
- (d) 2 Only

Q.84) With reference to India, the terms 'Chakma' and 'Hajong' pertain to

- a) Dance forms of Northwest India
- b) Sacred Groves site in North East India.
- c) Pre-historic cave paintings
- d) Communities from North East India

Q.85) In the context of Colonial India, Biku Patnaik, Ram Manohar Lohia, Sucheta Kriplani, J.P. Narayan are remembered as -

- a) Leaders of Quit India Movement
- b) Members of the Drafting Committee in the Constituent Assembly
- c) Members of the Interim Government in 1946
- d) Officers of the Indian National Army

Q.86) The term 'MARBURG' is talked about in the context of

- a) Development of India's own satellite navigation system
- b) Genes introduced in the genetically modified plants
- c) Type of virus which causes hemorrhagic fever
- d) Radio collar for wildlife tracking

Q.87) The SAMRIDH Scheme was recently in news in the affairs of:

- a) Promotion of green revolution in the eastern States
- b) Promotion of setting up of textile factories in rural areas
- c) Promotion of Indian start-ups
- d) None of the above

Q.88) It was declared as a national park in 1974. It is a world heritage site and famous for Great Indian one-horned rhinoceros. It has also been recognised as an Important Bird Area by BirdLife International for the conservation of avifaunal species. Recently, it became the first national park in India to be equipped with satellite phones.

Which of the following national parks is described in the above paragraph?

- a) Manas National Park
- b) Nameri National Park
- c) Raimona National Park
- d) Kaziranga National Park

Q.89) PM-DAKSH Scheme is being implemented by which of the following ministries?

- a) Ministry of Skill Development and Entrepreneurship
- b) Ministry of Social Justice and Empowerment
- c) Ministry of Minority Affairs
- d) Ministry of Tribal Affairs

Q.90) Which of the following statements with reference to the National Mission on Edible Oil-Oil Palm is correct?

- a) It proposes to cover an additional area of 6.5 million hectare for oil palm till the year 2025-26.
- b) Farmers will be provided assistance for planting materials of Rs. 29000 per hectare.
- c) It will have special focus on Lakshadweep Islands.
- d) None of the above

Q.91) Which of the following statements with reference to the National Legal Services Authority is correct?

1. It is a statutory body with the Chief Justice of India as its Executive Chairman.
2. It provides free legal services to women and children, members of scheduled caste and scheduled tribes.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2



d) Neither 1 nor 2

Q.92) Which of the following is the aim of the Ubharte Sitaare Fund launched recently?

- a) To provide debt and equity funding to export-oriented MSMEs
- b) To provide special funding to entrepreneurs aged below 21 years of age under the stand-up India scheme
- c) To support cottage industries of Aspirational districts
- d) To provide financial support to migrant workers establishing enterprise in their home district

Q.93) Consider the following statements with reference to Atlantic Meridional Overturning Circulation (AMOC):

- 1. It is a large system of ocean currents that is driven by differences in temperature and salinity.
- 2. Collapse of AMOC would lead to increased heating in the Northern Hemisphere.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.94) Consider the following statements with reference to Great Hornbill:

- 1. It is endemic to the Indian subcontinent.
- 2. It is the state bird of Nagaland, Arunachal Pradesh and Kerala.
- 3. It is listed as a critically endangered bird in the IUCN Red list.

Which of the above statements is/are incorrect?

- a) 1 and 3 only
- b) 2 only
- c) 3 only
- d) 1, 2 and 3

Q.95) Consider the following pairs of places in news and their location:

Places in News Country

- 1. Ningbo Port — Bangladesh
- 2. Cattle Island — Tanzania
- 3. Port-au-Prince — Haiti

Which of the above pairs is/are correctly matched?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 3 only

Q.96) Consider the following statements about e-RUPI:

- 1. E-rupi is a type of cryptocurrency.
- 2. It has been developed with the support of the Department of Financial Services and National Health Authority.
- 3. This Card is an Indian payment gateway that facilitates electronic payment at all Indian banks and financial institutions.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 3 only
- d) 2 only

Q.97) With reference to United Nations Security Council, consider the following statements:

- 1. A No vote from any of the five non permanent members blocks the passage of the resolution.
- 2. Any member of the UN which is not a member of the Security Council may vote in the discussion of any issue relating to them.
- 3. Peacekeeping Operations are an important role of the UNSC.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) All of the above.



Q.98) Consider the following statements about Nuclear Fusion:

1. High amounts of carbon is emitted during the Fusion reaction.
2. Fusion reactions do not occur naturally in nature.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.99) Consider the following statements:

1. Sunderbans was designated as a Biosphere Reserve by IUCN in 2001.
2. The Number of Kharai Camels is increasing in the Sundarban area.
3. Haliday Island Wildlife Sanctuary is a part of the Sundarbans Biosphere Reserve.

Which of the statements given above is/are correct?

- a) 2 and 3 only
- b) 1 and 3 only
- c) 3 only
- d) None of the above

Q.100) Consider the following statements:

1. One of the iconic paintings from the Bengal School is 'Bharat Mata'.
2. Raja Ravi Verma is one of the Painters of the Bengali School
3. Wash Technique is related to the Bengal School artists.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1 only



Q.1) Ans: C

Exp:

- The Pygmy Hog Conservation Programme is a collaboration among Durrell Wildlife Conservation Trust of UK, Assam Forest Department, Wild Pig Specialist Group of International Union for Conservation of Nature (IUCN) and the Ministry of Environment, Forest and Climate Change. The reintroduction programme began in 2008 with the Sonai-Rupai Wildlife Sanctuary, Orang National Park and Bornadi Wildlife Sanctuary, all of which are in Assam and now currently implemented by NGOs - Aaranyak and EcoSystems India.
- Pygmy hogs, the world's rarest and smallest wild pigs. It is listed as Endangered on the IUCN Red List and is designated as a Schedule I species in India under the Wildlife Protection Act of 1972.

Q.2) Ans: A

Exp:

- **Statement 1 is correct:** The Government has identified some world class iconic tourist stations in India to attract foreign and domestic tourists. Ministry of Tourism has identified 17 sites in 12 clusters in the country for development as Iconic Tourist Sites namely Taj Mahal & Fatehpur Sikri (Uttar Pradesh), Ajanta & Ellora (Maharashtra), Humayun's Tomb, Red Fort & Qutub Minar (Delhi), Colva (Goa), Amer Fort (Rajasthan), Somnath & Dholavira (Gujarat), Khajuraho (Madhya Pradesh), Hampi (Karnataka), Mahabalipuram (Tamil Nadu), **Kaziranga (Assam)**, Kumarakom (Kerala) and Mahabodhi Temple (Bihar).
- **Statement 2 is correct:** Four main types of vegetation exist in this park. These are alluvial inundated grasslands, alluvial savanna woodlands, tropical moist mixed deciduous forests, and tropical semi-evergreen forests. Based on Landsat data for 1986, percent coverage by vegetation is: tall grasses 41%, short grasses 11%, open jungle 29%, swamps 4%, rivers and water bodies 8%, and sand 6 %.
- **Statement 3 is correct:** The Important Bird Area (IBA) programme of Birdlife International aims to identify, monitor and protect a global network of IBAs for conservation of the world's birds and associated biodiversity. The IBAs serve as conservation areas for protection of birds at the global, regional or sub-regional level. Kaziranga

National Park has been designated the status of IBA due to its diversity of bird species.

- **Statement 4 is incorrect:** Kaziranga National Park is located in Golaghat and Nagaon, in the Karbi Anglong district of Assam in northeast India. Manas National Park is located in Chirang and Baksa District of Assam in northeast India and is contiguous with Royal Manas National Park of Bhutan.

Q.3) Ans: D

Exp:

- **Statement 1 is incorrect:** Conservation reserves and community reserves in India are terms denoting protected areas of India which typically act as buffer zones to or connectors and migration corridors between established national parks, wildlife sanctuaries and reserved and protected forests of India. The State Government may notify any community land or private land as Community Reserve, provided that the members of that community or individuals concerned are agreeable to offer such areas for protecting the fauna and flora, as well as their traditions, cultures and practice.
- **Statement 2 is incorrect:** No change in land use pattern shall be made within the Community Reserve, except in accordance with a resolution passed by the Management Committee and approval of the same by the State Government.
- **Statement 3 is correct:** There are 214 existing Community Reserves in India covering an area of 1302 km², which is 0.04% of the geographical area of the country (National Wildlife Database, Dec. 2020). Nagaland with 114 Community Reserves covering an area of 851.78 square kilometers has the maximum number in terms of area. It is about 3.9% of the total area of the state.

Q.4) Ans: A

Exp:

- Minister for Environment, Forest and Climate Change, releases Management Effectiveness Evaluation (MEE) of 146 National Park and Wildlife Sanctuaries in the country. At present, India has a network of 903 Protected Areas in the country covering about 5% of the total geographic area of the country. In order to assess the efficacy of Protected Areas, evaluation of management effectiveness is required.



- Management Effectiveness Evaluation (MEE) of Protected Areas (PAs) has emerged as a key tool for PA managers and is increasingly being used by governments and international bodies to understand the strengths and weaknesses of the protected area management systems
- MEE is a very important document that provides valuable guidance on various aspects of wildlife and protected areas to expand the MEE of Marine Protected Areas. A new framework for MEE of Marine Protected Areas has been also jointly prepared by WII and MoEF&CC and it will be a very useful document to implement.
- The Environment Minister also launched the Management Effectiveness Evaluation of Indian Zoos (MEE-ZOO) framework which proposes guidelines, criteria and indicators for evaluation of zoos of the country through Management Effectiveness Evaluation Process (MEE-ZOO) in a manner which is discrete, holistic and independent.

Q.5) Ans: A

Exp:

- The Directorate of Revenue Intelligence is the apex anti-smuggling agency of India, working under the Central Board of Indirect Taxes & Customs, Ministry of Finance, Government of India. It is tasked with detecting and curbing smuggling of contraband, including drug trafficking and illicit international trade in wildlife and environmentally sensitive items, as well as combating commercial frauds related to international trade and evasion of Customs duty.
- WCCB: Under Section 38 (Z) of the Wild Life (Protection) Act, 1972, it is mandated to collect and collate intelligence related to organized wildlife crime activities and to disseminate the same to State and other enforcement agencies.
- TRAFFIC-India works closely with the National and the State Governments and various agencies to help study, monitor and influence action to curb illegal wildlife trade and bring wildlife trade within sustainable levels
- WWF-India has been devotedly working to protect and secure natural heritage and ecology for more than 50 years.

Q. 6) Ans: C

Exp:

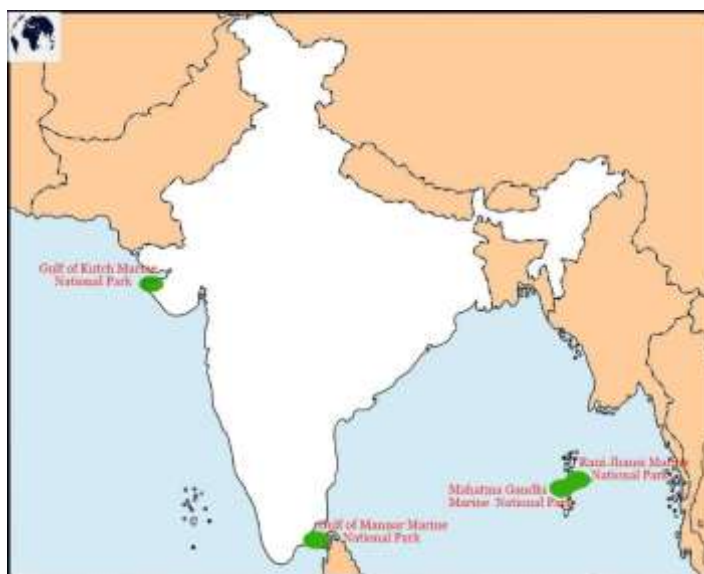
- Statement 1 is incorrect: The Conservation Reserve Management Committee is constituted under the wildlife (Protection) Act, 1972.
- Statement 2 is correct: The State Government shall constitute a conservation reserve management committee to advise the Chief Wildlife Warden to conserve, manage and maintain the conservation reserve.
- Statement 3 is incorrect: No change in the land use pattern shall be made within the community reserve, except in accordance with a resolution passed by the management committee and approval of the same by the State Government.

Q.7) Ans: B

Exp:

- **Gulf of Kutch Marine National Park:** Marine National Park in Gulf of Kutch is situated on the southern shore of the Gulf of Kutch in Jamnagar, Gujarat. It is the first national marine park of India with 42 islands on the Jamnagar coast in the Marine National Park.
- **Mahatma Gandhi Marine National Park:** Mahatma Gandhi Marine National Park is a national park of India on the Andaman Islands, situated 29 kms. from Port Blair. Most of the coral reefs in the park are fringing reefs and the park is an important breeding ground for turtles.
- **Gulf of Mannar Marine National Park:** Gulf of Mannar Marine National Park consists of 21 small islands and coral reefs in the Gulf of Mannar in the Indian Ocean of Tamil Nadu. The park has a high diversity of plants and animals in its marine and shore habitats. Dugong (Sea Cow), a vulnerable marine mammal, is also found here.
- **Rani Jhansi Marine National Park:** The Rani Jhansi Marine National Park (RJMNP) is located in the Andaman and Nicobar Islands in the Bay of Bengal and covers an area of 256.14 square kilometres. It was founded in 1996 and commemorates Lakshmibai, the Rani of Jhansi (1828-58). The park provides shelter to some very crucial wildlife and habitats which include Dugongs, Crocodiles, Marine Life, Coral Reefs, Mangroves, Terrestrial Moist forest and birds.

Please refer the following map for their relative locations:



Q.8) Ans: D

Exp:

- **Keetham lake**, also known as the **Sur Sarovar**, is a scenic lake just outside Agra. Alongside the Keetham Lake is the **Agra Bear Rescue Facility**, which is a Sloth bear rescue facility dedicated to rehabilitating previously enslaved 'dancing bears'. Originally created to supply water to the city of Agra in the summer, the wetland soon became an important and rich ecosystem. The site's patchwork of habitat types provides refuge to resident and migratory birds, and more than 60 species of fish. The site is important for bird species that migrate on the Central Asian Flyway, with over 30,000 waterbirds known to visit the reservoir annually.
- **Nawabganj Bird Sanctuary**: A shallow marshland 45 kilometres from Lucknow in Uttar Pradesh. Monsoon rains feed this diverse wetland while the Sarada Canal supplies additional water. The sanctuary supports recreation and tourism activities as well as local biodiversity. It is a haven for birds, with 25,000 waterbirds regularly recorded and 220 resident and migratory species documented.
- **Samaspur Bird Sanctuary**: A perennial lowland marsh typical of the Indo-Gangetic Plain in Raebareilly district. Its six connected lakes are heavily relevant to monsoon rains. Annual counts regularly find more than 75,000 birds present, with over 250 resident and migrant species documented.

- **Parvati Aranga Bird Sanctuary**: A permanent freshwater environment consisting of two oxbow lakes. These wetlands are characteristic of Uttar Pradesh and offer exceptional habitats for waterbirds, providing both roosting and breeding sites with over 100,000 birds documented in annual counts.

Q.9) Ans: D

Exp:

Sacred Groves:

- It refers to a piece of natural vegetation that is protected by a certain community due to religious reasons.
- Hunting and deforestation in these areas are banned but other activities with a more sustainable process such as honey and deadwood collection are allowed.
- **The Wild Life (Protection) Amendment Act, 2002**, provides government protection to these lands.
- The groves are treasure houses of rare species, and medicinal and economically important plants.

List of Sacred Groves of India:

State	Local term for Sacred Groves
Andhra Pradesh	Pavithravana
Arunachal Pradesh	Gumpa Forests (Attached to Buddhist monasteries)
Goa	Deorai, Pann
Jharkhand	Sarana
Karnataka	Devara Kadu
Kerala	Kavu, Sara Kavu
Maharashtra	Devrai, Devrahati, Devgudi

Manipur	Gamkhap, Mauhak (Sacred bamboo Reserve)
Meghalaya	Ki Law Lyngdoh, Ki Law Kyntang, Ki Law Niam
Orissa	Jahera, Thukuramma
Puducherry	Kovil Kadu
Rajasthan	Orans, Kenkris, Jogmaya
Tamil Nadu	Swami shoal, Koikadu
Uttarakhand	Deo Bhumi, Bugyal (Sacred Alpine Meadows)
West Bengal	Garamthan, Harithan, Jahera, Sabitrithan, Santalburithan

which also is a boundary between Nagarhole and Bandipur National Park

- **Silent Valley National Park:** It is the last remaining rain forest of Kerala. Silent Valley was declared a National Park in 1984 and formally inaugurated in 1985. The number of several common species of damselflies found in the region is alarmingly low. The national park is one of the last undisturbed tracts of South Western Ghats mountain rain forests and tropical moist evergreen forest in India. Plans for a hydroelectric project that threatened the park's biodiversity stimulated an environmentalist social movement in the 1970s, known as the Save Silent Valley movement, which resulted in cancelling the project and creating the park in 1980.
- **Banerghatta National Park:** It is located near Bangalore, Karnataka in the hills of the Anekal range. The park is part of a wildlife corridor for elephants which connects the BR Hills and the Sathyamangalam forest. The park is contiguous with Talli reserve forest in the southeast and Bilikal forest in the south. Recognition: It was declared as a National Park in 1974. In 2002 a portion of the park became a biological reserve, the Bannerghatta Biological Park. It is the first biological park in India to have a fenced forested elephant sanctuary. In 2006, India's first butterfly enclosure was inaugurated at the park.

Please refer the following map:

Q.10) Ans: D

Exp:

- **Bandipur National Park:** In 1974, Bandipur Tiger reserve was established as one of the first 9 Tiger Reserves under Project Tiger. In 1985, by including adjacent areas, it was enlarged and named Bandipur National Park. Location: Tiger Reserve is situated in two contiguous districts (Mysore and Chamarajanagar) of Karnataka. It is located in the tri-junction area of the States of Karnataka, Tamil Nadu and Kerala. Bandipur Tiger Reserve is surrounded by - Nagarhole Tiger Reserve (Karnataka) in the North West (Kabini Reservoir separates the two) Mudumalai Tiger Reserve (Tamil Nadu) in the South and Wayanad Wildlife Sanctuary (Kerala) in the South West.
- **Nagarhole National Park:** It is also known as Rajiv Gandhi National Park. It was established as a wildlife sanctuary in 1955 and was upgraded into a national park in 1988. It was declared as the 37th Tiger reserve under Project Tiger in 1999. The Park lies in the Western Ghats and is a part of the Nilgiri Biosphere Reserve. The Nagarhole River flows through the park, which joins the Kabini River



- The consortium of like-minded, regional and international organizations, created to oversee and coordinate conservation, campaigning and fundraising activities to help the plight of south Asia's vultures.
- **Objective:** To save three critically important species from extinction through a single programme.
- **SAVE partners:** Bombay Natural History Society, Bird Conservation Nepal, RSPB (UK), National Trust for Nature Conservation (Nepal), International Centre for Birds of Prey (UK) and Zoological Society of London.
- **Operation Save Kurma:** The operation was conducted to combat the proliferating illegal trade of live **turtles** and its parts from the country to destinations abroad.

Q.13) Ans: A

Exp:

- In 2009, the Environment and Coal Ministries had jointly placed the country's forested areas under two categories – Go and No-Go zones. The exercise is aimed at prioritising forest areas under the **Forest Conservation Act, 1980. Hence statement 2 is incorrect.**
- 'No Go' areas are those having either more than 10 per cent weighted forest cover (WFC) or more than 30 per cent gross forest cover (GFC). **Hence Statement 3 is incorrect.**
- 'No-Go' areas are regions that were classified by the Ministry of Environment and Forests and Climate Change as containing very dense forests and hence closed to coal mining. Diversion of forest land, which are rich in flora and fauna, for coal mining in these areas is the main cause of concern as they will have "avoidable serious adverse impact on forests and wildlife. If mining were to continue, even with afforestation and reclamation, it would not be possible to restore the region's biodiversity. **Hence statement 1 is correct.**

Q.14) Ans: A

Exp:

The All India Tiger Estimation done quadrennially is steered by the National Tiger Conservation Authority with technical backstopping from the Wildlife Institute

Q.11) Ans: B

Exp:

The Supreme Court order in **T N Godavarman Thirumalpad (1996) Case** identified 'deemed forests' as "land having the characteristic of forests irrespective of the ownership". This includes:

- Thickly wooded areas of the Revenue Department not handed over to the Forest Department.
- Thickly wooded areas recommended to be handed over to the Forest Department.
- Thickly wooded land distributed to grantees but not cultivated.
- Thickly wooded plantations of the Forest Department.

Hence option B is the most correct.

Q.12) Ans: C

Exp:

SAVE (Saving Asia's Vultures from Extinction):

of India and implemented by State Forest Departments and partners. The latest results of the 4th cycle conducted in 2018 had shown that India now has an estimated 2967. With this number, India is home to nearly 75% of the global tiger population and has already fulfilled its resolve of doubling tiger numbers, made at St. Petersburg in 2010, much before the target year of 2022. **Hence statement 4 is correct.**

4th cycle of all India Tiger Estimation- highlights:

- Highest number of tigers have been found in Madhya Pradesh (526), after that Karnataka has 524 and Uttarakhand is accommodating 442 tigers.
- In five years, the number of protected areas increased from 692 to over 860, and community reserves from 43 to over 100.
- While the 2014 census pegged the total number of striped big cats in the country at 2,226, the 2010 census put the figure at 1,706 and the 2006 version at 1,411, indicating that tiger numbers have been on the rise.
- While Pench Tiger Reserve in Madhya Pradesh recorded the highest number of tigers, Sathyamangalam Tiger Reserve in Tamil Nadu registered the "maximum improvement" in 2014. Hence statements 1 and 2 are correct.
- Chhattisgarh and Mizoram saw a decline in their tiger numbers while tiger numbers in Odisha remained constant. All other states witnessed a positive trend.
- Out of the 50 tiger reserves in the country, three reserves - Mizoram's Dampa reserve, West Bengal's Buxa reserve and Jharkhand's Palamau reserve - have no tigers left. As per the 2018 tiger estimation report, the number of tigers in Valmiki Tiger Reserve is pegged at 33. Hence statement 3 is incorrect.

Q.15) Ans: D

Exp:

- **Ranthambore National Park** is a national park in Rajasthan, named after the historic Ranthambore Fort which lies within the park. It is bounded to the north by the **Banas River** and to the south by the **Chambal river**. The Ranthambore National Park not only boasts of its populous tiger reserve, but also the heritage sites which includes the mighty Ranthambore Fort, **Jogi Mahal**, Ancient temple

and Rajbagh ruins which are the reminders of glory and grandeur of the Rulers of Ranthambore. Many **picturesque lakes** like Padam Talao, Malik Talao, Raj Bagh Talao and others enrich the landscape of this forest which consists of deciduous forest, small hills, valleys and lush greenery. Among many famous tigers, the **tigress Machli** was the most celebrated tiger in the world, which died in 2016.

- **Shivpuri National Park:** It is situated in Shivpuri District, Madhya Pradesh. It is a part of the upper Vindhyan hills. It has a diverse ecosystem consisting of lakes, dry deciduous & dry thorn forests. The forest is home to tigers, leopards, Nilgai, and Chinkara and Chousingha and Deers (Chital, Sambar and Barking Deer) among others.
- **Gandhi Sagar Wildlife Sanctuary:** Gandhi Sagar Sanctuary is a wildlife sanctuary situated on the northern boundary of Mandsaur and Nimach districts in Madhya Pradesh. The Chambal River passes through the sanctuary dividing it into two parts.
- **Ram Sagar Wildlife Sanctuary:** The Ramsagar wildlife sanctuary, Dholpur, is situated in the neighborhood of the famous Van Vihar wildlife sanctuary. Ramsagar Sanctuary holds beautiful Ramsagar Lake, which backs rich water life including freshwater crocodiles and various fishes and snakes.

Q.16) Ans: D

Exp:

- **Option 1 is incorrect:** Kaziranga National Park is also a wildlife sanctuary, a world heritage site, a tiger reserve as well as an elephant reserve. But **it is not a biosphere reserve.**
- **Option 2 is incorrect:** Panna National Park is also a wildlife sanctuary, a biosphere reserve and a tiger reserve under Project Tiger. But **it is neither a world heritage site nor an elephant reserve.**
- **Option 3 is incorrect:** Bandipur National Park is also a wildlife sanctuary and a part of Nilgiri biosphere reserve. But **it is neither a world heritage site nor an elephant reserve.**
- **Option 4 is correct:** Manas National Park or Manas Wildlife Sanctuary is a national park, UNESCO Natural World Heritage site, a Project Tiger reserve, an elephant reserve and a biosphere reserve. Located in the Himalayan foothills in



Assam, it is contiguous with the Royal Manas National Park in Bhutan. The park is known for its rare and endangered endemic wildlife such as the Assam roofed turtle, hispid hare, golden langur and pygmy hog. Manas is famous for its population of the wild water buffalo.

Q.17) Ans: B

Exp:

- **Option B is correct:** The overall aim of the India Ecodevelopment Project, approved in 1996, was to conserve biological diversity in seven globally significant protected areas (PAS) by implementing an eco-development strategy (prepared by the GOI). The strategy embraced a community based approach encouraging durable partnerships between Forest Department staff and local communities for access to and responsible use of forest resources.
- India Eco-development Project is being implemented in Seven Protected Areas in seven different States as an externally aided/Centrally Sponsored Plan Scheme under "Eco-development around Protected Areas including Tiger Reserves". It covers two National Parks and five Tiger Reserves. The main aim of the Project is to conserve bio-diversity through Eco-development, effective and extensive support for Eco-development and preparation of future Bio-diversity Projects. The project is being implemented in seven areas namely Buxa, Palamau, Nagarhole, Periyar, Pench, Ranthambore Tiger Reserves and Gir National Park.

Q.18) Ans: C

Exp:

- The Global Tiger Initiative (GTI) was launched in 2008 as a global alliance of governments, international organizations, civil society, the conservation and scientific communities and the private sector, with the aim of working together to save wild tigers from extinction. In 2013, the scope was broadened to include Snow Leopards.
- The Global Tiger Initiative (GTI) program of the World Bank brought global partners together to strengthen the tiger agenda. Over the years, the initiative has been institutionalized as a separate entity in the form of the Global Tiger Initiative Council (GTIC) with the Global Tiger Forum as one of its arms.

- The World Bank hosted the GTI Secretariat until July 1 2015 and acted as a convener of a global network working to save wild tigers and snow leopards from extinction. The Secretariat helped with the strategic imperatives agreed to in the St. Petersburg Declaration on Tiger Conservation and the Global Tiger Recovery Program.

Q.19) Ans: C

Exp:

- **Statement 1 is correct:** The Graded Action Plan was notified in 2017 for Delhi and the National Capital Region (NCR). These are institutionalized measures to be taken when air quality deteriorates, hence works only as an emergency measure. The plan does not include action by various state governments to be taken throughout the year to tackle industrial, vehicular and combustion emissions.
- **Statement 2 is correct:** GRAP includes the measures which will be taken by different government agencies to prevent worsening of Air Quality of Delhi-NCR and prevent PM10 and PM2.5 levels to go beyond the 'moderate' national Air Quality Index (AQI) category. If air quality reaches the severe + stage, GRAP talks about shutting down schools and implementing the odd-even road-space rationing scheme. When the air quality shifts from poor to very poor, the measures listed have to be followed since the plan is **incremental in nature**.
- **Statement 3 is incorrect:** The plan requires action and coordination among 13 different agencies in Delhi, Uttar Pradesh, Haryana and Rajasthan (NCR areas). At the head of the table is the EPCA, mandated by the Supreme Court. Before the imposition of any measures, EPCA holds a meeting with representatives from all NCR states, and a call is taken on which actions have to be made applicable in which town.

Q.20) Ans: D

Exp:

- **Statement 1 is incorrect:** When forest is diverted for non-forest use, the developer pays for **compensatory afforestation**. In addition, the government charges the developer, NPV, to compensate the forest's lost ecosystem services till the afforested area starts providing comparable benefits. **Net Present Value (NPV)** is a mandatory one-time payment that a user has to make for



diverting forestland for non-forest use, under the Forest (Conservation) Act, 1980.

- **Statement 2 is incorrect:** The Net Present Value is calculated on the basis of the services and ecological value of the forests. It depends on the location and nature of the forest and the type of industrial enterprise that will replace a particular parcel of forest. These payments go to the Compensatory Afforestation Fund (CAF) and are used for afforestation and reforestation. The CAF is managed by the Compensatory Afforestation Management and Planning Authority (CAMPA).
- **Statement 3 is incorrect:** The Forest Advisory Committee constituted by the Ministry of Environment, Forest and Climate Change decides on whether forests can be diverted for projects and the NPV to be charged. However, some projects have been given exemption from paying the Net Present Value like construction of Schools, Hospitals, village tanks, laying down of optical fibre etc. Whereas, Projects like underground mining and wind energy plants have been given a 50% exemption from NPV.

Q.21) Ans: A

Exp:

- **Statement 1 is incorrect:** The Government of India had set a **target of 175 GW renewable power** installed capacity by the end of 2022. This includes 60 GW from wind power, 100 GW from solar power, 10 GW from biomass power and 5 GW from small hydro power. The renewable energy capacity in India is **currently 136 Giga Watts**, which is about 36% of its total energy capacity. The 2030 renewable targets were announced to be 450 gigawatt.
- **Statement 2 is incorrect:** **UJALA(Unnat Jyoti by Affordable LEDs for All)** scheme's main objective is to promote efficient lighting, enhance awareness on using efficient equipment which reduces electricity bills and help conserve the environment. UJALA is a **zero-subsidy scheme** launched by the Government in 2015. It is touted as the world's largest domestic lighting project. It is implemented by Energy Efficiency Services Limited (EESL), a joint venture of PSUs under the Ministry of Power.
- **Statement 3 is correct:** One of the primary characteristics of a smart grid is its ability to self-heal. Self-healing capabilities minimize blackouts because they allow for continuous self-assessments that inspect, analyze, react to, and automatically respond to problems. Smart energy

systems rely on active consumer participation. Consumers have to accept, install, and use smart products and services. They need to provide data and adjust their behavior.

Q.22) Ans: B

Exp:

- **Statement 1 is correct:** Adaptation Fund was established under the Kyoto Protocol of the UN Framework Convention on Climate Change. It finances projects and programmes that help vulnerable communities in developing countries adapt to climate change. Initiatives are based on country needs, views and priorities.
- **Statement 2 is correct:** The Clean Development Mechanism(CDM) is a source of income for the UNFCCC Adaptation Fund, which was established to finance adaptation projects and programmes in developing country Parties to the Kyoto Protocol that are particularly vulnerable to the adverse effects of climate change. The Adaptation Fund is financed by a 2% levy on CERs issued by the CDM. The Fund is also financed by government and private donors.
- **Statement 3 is incorrect:** The Fund is supervised and managed by the Adaptation Fund Board (AFB). The AFB is composed of 16 members and 16 alternates and meets at least twice a year. The World Bank serves as trustee of the Adaptation Fund on an interim basis.

Q.23) Ans: C

Exp:

- **Statement 1 is correct:** Gasification is a technological process that can convert any carbonaceous (carbon-based) raw material such as coal into fuel gas, also known as synthesis gas (syngas for short). It occurs generally in a high temperature/pressure vessel where oxygen (or air) and steam are directly contacted with the coal or other feed material causing a series of chemical reactions to occur that convert the feed to syngas and ash/slag (mineral residues).
- **Statement 2 is correct:** Syngas is so called because of its history as an intermediate in the production of synthetic natural gas. Syngas is a mixture consisting primarily of methane (CH₄), carbon monoxide (CO), hydrogen (H₂), carbon dioxide (CO₂) and water vapour (H₂O). Syngas can



be used to produce a wide range of fertilizers, fuels, solvent and synthetic materials.

feasible, curb greenhouse gas emissions associated with growing food.

Q.24) Ans: A

Exp:

- **Statement 1 is incorrect:** The removal of Sulfur Dioxide is called Flue-gas Desulphurization (FGD). It seeks to remove gaseous pollutants viz. SO₂ from exhaust flue gases generated in furnaces, boilers, and other industrial processes due to thermal processing, treatment, and combustion.
- **Statement 2 is correct:** Lime plays a key role in many air pollution control applications. Lime is used to remove acidic gases, particularly sulfur dioxide (SO₂) and hydrogen chloride (HCl), from flue gases. Lime-based technology is also being evaluated for the removal of mercury. Lime is more reactive than limestone, and requires less capital equipment. There are two main methods for cleaning flue gases from coal combustion at electric generating stations: dry scrubbing and wet scrubbing. Lime is used in both systems.

Q.25) Ans: C

Exp:

- **Statement 1 is incorrect:** **Permaculture** is an approach to land management and philosophy that focuses on designing human systems based on natural ecosystems. It brings production of food closer to consumers and the consumer's wastes back into the cycle. It is an important component of Climate Smart Agriculture (CSA). Whereas, **Monocropping** is the practice of growing the same crop on the same plot of land, year after year. This practice depletes the soil of nutrients (making the soil less productive over time), reduces organic matter in soil and can cause significant erosion. Monocropping is more vulnerable to climate vagaries as compared to Permaculture.
- **Statement 2 is correct:** Conservation tillage practices reduce erosion by protecting the soil surface and allowing water to infiltrate instead of running off. This also increases water infiltration by as much as 30 to 45% compared to conventional tillage systems. This means less water is running off fields and into waterways, which may carry agrochemicals.
- **Statement 3 is correct:** The Climate-smart Agriculture aims to: Improve farmer productivity, and as a result, livelihoods; make farms more resilient to climate impacts they're facing now, and to those likely to hit in the future; and, where

Q.26) Ans: A

Exp:

- **Statement 1 is incorrect:** The Stockholm Convention on Persistent Organic Pollutants is a global treaty to protect human health and the environment from chemicals that remain intact in the environment for long periods. It aims to eliminate or restrict the production and use of persistent organic pollutants (POPs). India had ratified the Stockholm Convention on January 13, 2006.
- **Statement 2 is correct:** The Global Environmental Facility (GEF) is the designated interim financial mechanism for the Stockholm Convention. The GEF is jointly managed by the United Nations Development Programme (UNDP), the World Bank, and the United Nations Environment Programme (UNEP). The financial mechanism was established to help tackle our planet's most pressing environmental problems. It provides funds to the developing countries and transition economies for projects related to climate change, biodiversity, the ozone layer, etc.
- **Statement 3 is correct:** The Stockholm convention calls to **ban nine of the dirty dozen chemicals (key POPs), limit the use of DDT to malaria control, and curtail inadvertent production of dioxins and furans**. The convention listed twelve distinct chemicals in three categories:
 - Eight pesticides (aldrin, chlordane, DDT, dieldrin, endrin, heptachlor, mirex and toxaphene)
 - Two industrial chemicals (poly chlorinated biphenyls and hexachlorobenzene)
 - Two unintended by-products of many industrial processes involving chlorine (poly chlorinated dibenzo-p-dioxins and dibenzofurans, commonly referred to as dioxins and furans).

Q.27) Ans: D

Exp:

- **Statement 1 is incorrect:** 'Green' crackers have a small shell size compared to traditional crackers. They are produced using less harmful raw materials and have additives which reduce emissions by suppressing dust. The green crackers are also supposed to have formulations

that substitute some of the raw materials to reduce PM 2.5 particles, sulphur dioxide and nitrogen dioxide emissions as well.

- **Statement 2 is incorrect:** Green crackers don't contain banned chemicals such as lithium, arsenic, barium and lead. Examples of Green Crackers - **Safe Water Releaser (SWAS)**, **Safe Thermite Cracker (STAR)** and **Safe Minimal Aluminium (SAFAL)**. In Green Crackers, commonly used polluting chemicals like aluminum, barium, potassium nitrate and carbon are removed or sharply reduced so that the emissions come down by 15 to 30%.
- **Statement 3 is correct:** Green crackers release water vapour and don't allow the dust particles to rise. They are designed to have 30% less particulate matter pollution and are not completely environment-friendly. These are firecrackers made with a reduced shell size, and produce 30% to 35% less emissions compared to normal firecrackers. The green crackers also reportedly emit 125 decibels of sound, compared to 160 decibels emitted by non-green variants.

Q.28) Ans: D

Exp:

- **Statement 1 is correct:** Solar radiation management (SRM or solar geoengineering) is a theoretical approach for reducing some of the impacts of climate change. If ever done, it would involve reflecting a small amount of sunlight back out into space. It involves geoengineering technologies that aim to reduce the inflow of solar energy, rather than to reduce global warming by reducing greenhouse gases.
- **Statement 2 is correct:** Whitening clouds, injecting particles into the stratosphere (Stratospheric aerosol injection) , or putting sunshades in space could increase Earth's reflectivity, thereby reducing incoming solar radiation and offsetting some of the warming associated with increasing GHG concentrations.
- **Statement 3 is correct:** Clouds, especially, low, thick clouds, reflect incoming sunlight (shortwave radiation) back to space. High clouds reflect less solar radiation and trap longwave infrared radiation (heat) re-radiated from Earth's surface. Thick clouds (such as stratocumulus) reflect a large amount of incoming solar radiation, translating to a high albedo. Thin clouds (such as cirrus) tend to

transmit more solar radiation and, therefore, have a low albedo.

Q.29) Ans: B

Exp:

- **Statement 1 is correct:** Integrating animals into crop production systems yields considerable benefits in **improved soil health, reduced risks associated with raising a single product, reductions in fertilizer input and animal feed costs, reduced labor and machinery costs, and increased carbon sequestration**. Grazing cropland improves soil fertility by increasing soil microbial density and organic matter due to the addition of manure. It can also provide significant benefits for farmers who use cover crops and no-till methods as the animals can graze the cover crops while lightly integrating their manure into the soil with their hooves.
- **Statement 2 is incorrect:** Conventional tilling, that is plowing up the dirt, encourages microbial action. That can lead to CO2 emissions and the more potent greenhouse gases methane and nitrous oxide. Whereas, No-till can mitigate the greenhouse gases that contribute to climate change. Adopting no-till management has many benefits for sustainable management of soils, such as improving soil structure, reducing erosion, enhancing soil moisture, and Carbon storage. There is more Carbon in the soil with no-till management compared to full tillage management, and therefore less Carbon in the atmosphere.
- **Statement 3 is correct:** **The rise in meat consumption and production causes considerable environmental damage including deforestation, water contamination and soil degradation**. High meat consumption has also largely led to industrialized agriculture practices that have been criticized for the use of antibiotics and hormones, and increasing risks in food safety and animal welfare. Decreasing meat production, primarily of ruminants, reduces water use, soil degradation, pressure on forests, and manure and pollution into water systems. Reducing demand of meat by a relatively small amount would have a significant absolute impact on GHG emissions, human health and the environment associated with livestock production.

Q.30) Ans: A

Exp:

- **Statement 1 is correct:** Rainmaking, also known as artificial precipitation, artificial rainfall and pluvioculture, is the act of attempting to artificially induce or increase precipitation, usually to stave off drought or the wider global warming. According to the clouds' different physical properties, this can be done using airplanes or rockets to sow to the clouds with catalysts such as dry ice, silver iodide and salt powder, to make clouds rain or increase precipitation, to remove or mitigate farmland drought, to increase reservoir irrigation water or to solve the global warming problem.
- **Statement 2 is incorrect:** Cloud seeding and Artificial Rain can only be done if the meteorological conditions for the are suitable, and if the moisture content in the local atmosphere meets the requisite criteria. However, it can only provide immediate symptomatic relief in terms of suppressing the pollutants present in the air. Ultimately, if the sources of the pollutants, i.e. vehicles, industry and construction are allowed to emit unabated, artificial rains through cloud seeding will have only a limited and temporary impact.

Q.31) Ans: C

Exp:

- **Statement 1 is incorrect:** Kigali Amendment has divided the signatory parties into three groups-
 - Group I - rich and developed economies like the USA, UK and EU countries who will start to phase down HFCs by 2019 and reduce it to 15% of 2012 levels by 2036.
 - Group II -emerging economies like China, Brazil as well as some African countries who will phase down by 2024 and reduce it to 20% of 2021 levels by 2045.
 - Group III - developing economies and some of the hottest climate countries like India, Pakistan, Iran, Saudi Arabia who will start phasing down HFCs by 2028 and reduce it to 15% of 2024-2026 levels till 2047.
- **Statement 2 is correct:** HFCs are refrigerants used in air conditioning and refrigeration systems. They are **potent greenhouse gases with global warming potential** (GWP) thousand times more than carbon dioxide (CO₂). The Montreal Protocol was signed to eliminate the use of chemicals that caused the ozone hole. However, the replacement for the

ozone-depleting chemicals was HFCs, which caused global warming. At the Kigali meeting, the world agreed to reduce the use of HFCs and move to refrigerants that neither cause global warming nor create an ozone hole.

- **Statement 3 is correct:** Kigali Amendment strengthens the Paris Agreement which sets an ambitious target of restricting the rise in global temperature below 20 Celsius, as compared to pre-industrial level. Unlike the Paris agreement, Kigali Amendment to the Montreal Protocol is legally binding with mandatory HFC reduction targets for countries. It gives clear, concrete and mandatory targets with fixed timelines to the signatory parties to achieve their targets.

Q.32) Ans: C

Exp:

- **Statement 1 is correct:** The term carbon negative takes the concept of carbon neutrality a step further. Where carbon neutral is an equal balance between the carbon dioxide you emit into the atmosphere and how much you remove from it, **carbon negative means you remove more carbon dioxide from the atmosphere than you emit**. The government of **Bhutan is the only country** in the world to make such a switch and the first country to become carbon negative.
- **Statement 2 is incorrect:** Carbon neutrality, or having a net zero carbon footprint, refers to achieving net zero carbon emissions by balancing a measured amount of carbon released with an equivalent amount sequestered or offset, or buying enough carbon credits to make up the difference.
- **Statement 3 is incorrect:** **Greenwashing is the process of conveying a false impression or providing misleading information about how a company's products are more environmentally friendly.** Companies apply greenwashing techniques to the promotion of their products in order to appeal to the environmentally-conscious consumer so that they can gain market share and a competitive advantage over their rivals. It is a deceitful advertising gimmick.

Q.33) Ans: B

Exp:

- **Statement 1 is correct:** The **Paris Agreement** is an agreement within the United Nations Framework Convention on Climate Change (**UNFCCC**), on climate change mitigation, adaptation, and

finance, signed in 2016. As of March 2021, 191 members of the UNFCCC are parties to the agreement. The Paris Agreement sets out a global framework to avoid dangerous climate change by limiting global warming to well below 2°C and pursuing efforts to limit it to 1.5°C. It also aims to strengthen countries' ability to deal with the impacts of climate change and support them in their efforts.

- **Statement 2 is incorrect:** The Paris Agreement replaced the Kyoto Protocol. Unlike the Kyoto Protocol, which established top-down legally binding emissions reduction targets (as well as penalties for noncompliance) for developed nations only, the Paris Agreement requires that all countries—rich, poor, developed, and developing—do their part and slash greenhouse gas emissions to solve the global problem of climate change through Nationally determined contributions (NDCs).
- **Statement 3 is incorrect:** As of July 2021, 191 members of the United Nations Framework Convention on Climate Change (UNFCCC) are parties to the agreement. Iran, Iraq, Libya, Yemen and Eritrea have not ratified the agreement. South Sudan, was the latest nation to ratify the deal on 23 February 2021.

Q.34) Ans: D

Exp:

- **Pair 1 is incorrect:** REDD refers to reducing emissions from deforestation and forest degradation; REDD+ refers to the framework created by the UNFCCC Conference of the Parties (COP) to guide activities in the forest sector for the conservation of forest carbon stocks, sustainable management of forests, and enhancement of forest carbon stocks.
- **Pair 2 is incorrect:** The Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity is an international agreement which aims at sharing the benefits arising from the utilization of genetic resources in a fair and equitable way. It was adopted in 2010 in Nagoya, Aichi Province, Japan, & entered into force in 2014. The protocol is legally binding & open to only CBD ratified countries (excludes USA, Andorra). It does not apply to Human Genetic Material. Nagoya protocol's strategic plan with 20 targets is called "Aichi Target". 'The Access & Benefit Sharing

(ABS) Clearing House' is a key tool which facilitates the smooth implementation of the Nagoya Protocol.

- **Pair 3 is incorrect:** The Talanoa Dialogue is a process which would help countries implement their Intended Nationally Determined Contributions by the end of 2020. The Talanoa Dialogue was launched at COP 23 under the Presidency of the Republic of Fiji. It is a facilitative process through which the deadlock regarding climate change can be brought together by drawing participants to share their stories and sidestepping the power struggle.

Q.35) Ans: D

Exp:

- **Statement 1 is correct:** The Water Act is enacted with the objective of prevention & control of pollution in India. The Act aims at the maintaining or restoring the wholesome nature of water for the establishment of Boards and to vest them with such powers so as to enable them to carry out the purposes of the Act. The Water (Prevention & Control of Pollution) Act, 1974, deals with the constitution of the Central Board, State Boards, and the Functions of Pollution Control Board.
- **Statement 2 is correct:** According to Section 4 of the Water Act, 1974, the State Boards must be constituted by the State Government, which is known as the State Pollution Control Board. Every State Board must be considered as a body corporate having perpetual succession with the power to hold, acquire, and dispose of the property. It can also enter into a contract with any person or party. They can sue or can be sued in the name of State Boards.
- **Statement 3 is correct:** State Pollution Control Board has to inspect sewage or the trade effluents works and plants to treat sewage and trade effluents. They are also required to review plans, specifications, or other data relating to plants set up for the works for the purification, treatment of water, and the system of the disposal of sewage or trade effluents connected with the grant of any consent.

Q.36) Ans: D

Exp:

- **Statement 1 is correct:** Fuel Cell Electric Vehicle (FCEV) engines are similar to the conventional

internal combustion engines because they also rely on a constant supply of fuel (hydrogen) and oxygen. Unlike a battery-electricity vehicle, it does not store energy.

- **Statement 2 is correct:** FCEVs do not need to be plugged in for charging, like battery-powered EVs. They are filled up with a nozzle, just like in a petrol or diesel station. Their fuel cells are recharged by refilling with hydrogen, which can take as little as 5 minutes at a filling station.
- **Statement 3 is correct:** Fuel cells that use pure hydrogen fuel are completely carbon-free. They are more efficient than conventional internal combustion engine vehicles and produce no tailpipe emissions—they only emit water vapor and warm air.

Q.37) Ans: B

Exp:

Indian Council of Agricultural Research (ICAR), Ministry of Agriculture and Farmers Welfare, Government of India launched a flagship network project 'National Innovations in Climate Resilient Agriculture' (NICRA) in 2011. The project aims at strategic research on adaptation and mitigation, demonstration of technologies on farmers' fields and creating awareness among farmers and other stakeholders to minimize the climatic change impacts on agriculture.

Q.38) Ans: D

Exp:

- The Pensilungpa Glacier located in Ladakh's Zaskar Valley is retreating due to increase in temperature and decrease in precipitation during winters, a recent study has found. Since 2015, the Wadia Institute of Himalayan Geology (WIHG) at Dehradun, an autonomous body under the Department of Science and Technology, has been working on various aspects on glaciology – glacier health (mass balance) monitoring, dynamics, discharge, past climatic conditions, speculation for future climate change and its impact on glaciers in this region.
- A glacier is a huge mass of ice that moves slowly over land. The term "glacier" comes from the French word glace (glah-SAY), which means ice. Glaciers are often called "rivers of ice." Glaciers fall into two groups: alpine glaciers and ice sheets. Alpine glaciers form on mountainsides and move downward through valleys

Q.39) Ans: D

Exp:

- The Union Ministry of Minority Affairs implements programmes/ schemes for socio-economic and educational empowerment of the six (06) centrally notified minority communities namely Muslims, Christians, Sikhs, Buddhists, Parsis and Jains. These schemes/programmes implemented by the Ministry during last three years in brief are as under:-
 - Pre-Matric Scholarship Scheme, Post-Matric Scholarship Scheme, Merit-cum-Means based Scholarship Scheme - For educational empowerment of students through Direct Benefit Transfer (DBT) mode.
 - Maulana Azad National Fellowship Scheme - Provide fellowships in the form of financial assistance.
 - Naya Savera - Free Coaching and Allied Scheme - The Scheme aims to provide free coaching to students/candidates belonging to minority communities for qualifying in entrance examinations of technical/ medical professional courses and various competitive examinations.
 - Padho Pardesh - Scheme of interest subsidy to students of minority communities on educational loans for overseas higher studies.
 - Nai Udaan - Support for students clearing Prelims conducted by Union Public Service Commission (UPSC), State Public Service Commission (PSC) Staff Selection Commission (SSC) etc.
 - Nai Roshni - Leadership development of women belonging to minority communities
 - Seekho Aur Kamao - Skill development scheme for youth of 14 - 35 years age group and aiming at providing employment and employment opportunities, improving the employability of existing workers, school dropouts etc
 - Pradhan Mantri Jan Vikas Karyakram (PMJKV) - Being implemented to provide basic infrastructure such as Schools, Colleges, ITIs, Polytechnics, Hostels, Sadbhav Mandaps, Skill Development Centres, Drinking water and Sanitation facilities, Primary Health Centres etc. in deprived areas of the Country.
 - Jiyo Parsi - Scheme for containing population decline of Parsis in India. USTTAD (Upgrading the Skills and Training in Traditional Arts/Crafts for Development). Hunar Haats are being organised across the country to provide

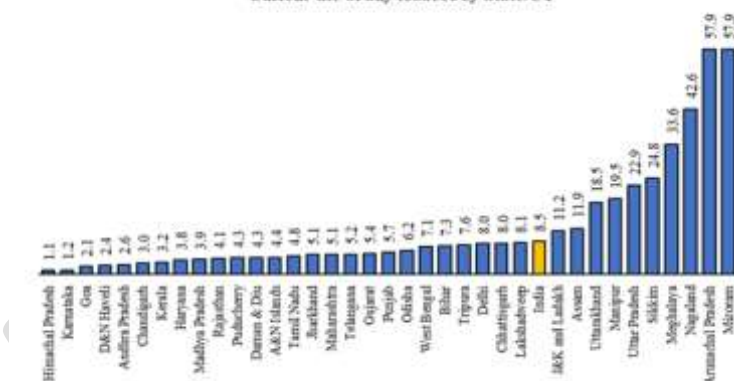
employment opportunities and markets to artisans/craftsmen.

- Nai Manzil - A scheme for formal school education & skilling of school dropouts.
- Hamari Dharohar- A scheme to preserve the rich heritage of minority communities of India under the overall concept of Indian culture.
- Maulana Azad Education Foundation (MAEF) implements education and skill related schemes as follows:- (a) Begum Hazrat Mahal National Scholarship for Meritorious Girls belonging to the economically weaker sections of Minorities (b) Gharib Nawaz Self Employment Scheme started in 2017-18 for providing short term job oriented skill development training to youth (c) Bridge Course for madrasa students & school dropouts.
- Equity to National Minorities Development and Finance Corporation (NMDFC) for providing concessional loans to minorities for self-employment and income generating ventures.

private). The first three rounds of GYTS were conducted in 2003, 2006 and 2009.

- **Statement 3 is incorrect:** The current use of tobacco among students across the States/ UTs ranged from the highest in Arunachal Pradesh and Mizoram (58% each) to the lowest in Himachal Pradesh (1.1%) and Karnataka (1.2%).

Current use of any tobacco by State/UT



Q.41) Ans: B

Exp:

- Janaushadhi Sugam App has been developed by the Bureau of Pharma PSUs of India (BPPI) under the Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers. Amid nationwide lockdown due to COVID-19 crisis, Janaushadhi Sugam Mobile App is helping people in a big way to locate their nearest Pradhan Mantri Bharatiya Janaushadhi Kendra (PMJAK) and availability of affordable generic medicine with its price.

It is digital platform, by the virtue of which one can avail a host of user-friendly options like locate nearby Janaushadhi kendra through direction guided by Google Map for location of the Janaushadhi kendra, search Janaushadhi generic medicines, analyze product comparison of Generic and Branded medicine in form of MRP & overall Savings, etc. Janaushadhi Sugam Mobile App is available on both Android & I-phone platforms. It can be downloaded free of cost by the user from Google Play Store and Apple Store.

Q.42) Ans: A

Exp:

- The Asiatic Lion Reintroduction Project is an initiative of the Indian Government to provide safeguards to the Asiatic lion from extinction in the wild by means of reintroduction. The programme has been launched for the conservation of the Asiatic Lion, whose last remaining wild population is in Gujarat's Asiatic Lion Landscape.

Q.40) Ans: B

Exp:

- **Statement 1 is correct :** Between the last two surveys, the current use declined by 42% (2009-2019). Nearly one-fifth of the students aged 13-15 used any form of the tobacco product (smoking, smokeless, and any other form) in their life.

The other findings of the report are as follows:

- Prevalence of tobacco use among boys was 9.6% and among girls was 7.4%.
- The prevalence of smoking tobacco was 7.3%. In the case of smokeless tobacco products, the prevalence was 4.1%.
- The use of e-cigarette among the students was 2.8%.
- Use of any form of tobacco was higher among boys than girls.
- **Statement 2 is incorrect:** The fourth round of Global Youth Tobacco Survey (GYTS-4) was conducted in 2019 by the International Institute for Population Sciences (IIPS) under the Ministry of Health and Family Welfare (MoHFW). The survey was designed to produce national estimates of tobacco use among school going children aged 13-15 years at the state level and Union Territory (UT) by sex, location of school (rural-urban), and management of school (public-



- Six new sites apart from the Kuno-Palpur Wildlife Sanctuary (As of now finalized for Cheetah reintroduction) have been identified under Project Lion. The Wildlife Institute of India, along with the Gujarat Forest Department, had created a Project Lion proposal and sent it to the Union Ministry of Environment, Forest and Climate Change.
- The six new sites identified for possible lion relocation in the future include:
 - Madhav National Park, Madhya Pradesh
 - Sitamata Wildlife Sanctuary, Rajasthan
 - Mukundra Hills Tiger Reserve, Rajasthan
 - Gandhi Sagar Wildlife Sanctuary, Madhya Pradesh
 - Kumbhalgarh Wildlife Sanctuary, Rajasthan
 - Jessore-Balaram Ambaji WLS and adjoining landscape, Gujarat

Q.43) Ans: b

Exp:

- **Biodiversity hotspots** is a biogeographic region that is both a significant reservoir of biodiversity and is threatened with destruction. **Hence Statement 1 is Incorrect**
 - The term 'Biodiversity Hotspot' was coined by *Norman Myers*
 - "Biodiversity hotspots" can also be defined as the regions which are known for their high species richness and endemism
- **Criteria for biodiversity hotspot:** According to Conservation International, a region must meet two strict criteria:
 - It must contain at least 1,500 species of vascular plants (> 0.5% of the world's total) as endemic – which is to say, it must have a high percentage of plant life found nowhere else on the planet. A hotspot, in other words, is irreplaceable.
 - It has to have lost at least 70% of its original habitat. (It must have 30% or less of its original natural vegetation)
 - **Hence Statement 2 & 3 are correct**
- There are major four biodiversity hotspots in India:
 - The Himalayas
 - Indo-Burma Region
 - The Western Ghats
 - Sundaland

- **Hence Statement 4 is correct**

Q.44) Ans: d

Exp:

- A **community** is defined as a group of individuals of different species, living within a certain geographical area
 - In other terms, Community is a diverse group of organisms that interact in a common location. For instance, a forest, inhabited by animals and various species of plants, as well as soil bacteria and fungi constitutes a biological community.
- **Statement 1 is Correct** : species evenness refers to how close in numbers each species in an environment is. Mathematically, it is a measure of biodiversity which quantifies how equal the community is numerically. Say, if there are 5 dogs and 20 cats then the community isn't even
- **Statement 2 is Correct**: Species richness is the measure of number of species found in a community
- **Statement 3 is Correct** : Low species evenness means there is large inequality in numbers between different species. High richness means there are many different species. Hence few species would dominate the region of their inhabitation

Q.45) Ans: d

Exp:

- An Invasive species is a non-indigenous species that spreads from the point of introduction and becomes abundant. It can be any kind of living organism that is not native to an ecosystem. And causes harm. **Hence statement a is Correct**
- To be Invasive, a species must adapt to the new area easily giving them an edge. It must reproduce quickly. It must harm property, the economy or the native plants and animals of the region
- Not all Non-native species are Invasive. For example, most of the food crops grown in the United States, including popular varieties of wheat, tomatoes and rice are not native to the region. **Hence Statement b is Correct**
- India has adopted Aichi Biodiversity Target 9 as its National Biodiversity Target 4 - i.e. to identify invasive alien species and their pathways of introduction, and to develop strategies to manage



prioritized invasive alien species by 2020. **Hence Statement c is Correct**

- **Statement d is Incorrect** : The Ministry of Environment and Forests issues approval along with quarantine certificates for the export of wild animals and articles under the Wildlife (Protection) Act 1972. Also, The Destructive Insects and Pests Act 1914 aims to prevent introductions into India, and the transport from one province to another, of any fungus or other pests which is, or may be destructive to crops

Q.46) Ans: c

Exp:

- **Alpha diversity** represents the species richness in an ecosystem. It is measured by counting the number of taxa (usually species) within a particular area, community or ecosystem. **Hence Statement (ii)**
- **Beta diversity** is the diversity of species between two habitats or regions or, it's the ratio between Regional and local species Diversity. In other words, species diversity between two adjacent ecosystems and is obtained by comparing the number of species unique to each of the ecosystem. **Hence Statement (i)**
- **Gamma Diversity** is a measure of overall diversity for different ecosystems with a region with characteristic flora & fauna bounded by natural boundaries or in other words, It refers to the diversity of the habitats over the total landscape or geographical area. **Hence Statement (iii)**

Q.47) Ans: a

Exp:

- **Statement 1 is Correct** : Closely-related species have in common much of their hereditary characteristics. For instance, about 98.4 percent of the genes of humans and chimpanzees are the same. Small populations lose genetic variability because of genetic drift, and inbreeding within populations can further decrease individual variability.
- **Statement 2 is Correct** : Lower genetic variation depresses individual fitness, resistance to disease and parasites, and flexibility in coping with environmental challenge
- **Statement 3 is Incorrect** : Genetic diversity occurs within a species as well as between species. For example, poodles, German shepherds and golden retrievers are all dogs, but they all look different.

Q.48) Ans: b

Exp:

- **Statement 1 is Incorrect** : For Species Diversity, 'Zero' would mean infinite diversity, and 'one' represents only one species present.
- **Statement 2 is Correct** : In general, species diversity decreases as we move away from the equator towards the poles.
 - This can be understood from the following factors : (i) Temperature decreases and conditions become harsh (ii) Both the amount and intensity of solar radiation decreases (iii) Vegetation decreases. (iv) Less resources available to support species.
 - Speciation is generally a function of time and environmental stability, so if conditions are too harsh, it is difficult for the species to survive and adapt. This results in decrease in biodiversity towards the poles
- **Statement 3 is Incorrect** : With very few exceptions, tropics (latitudinal range of 23.5° N to 23.5° S) harbour more species than temperate or polar areas. The higher temperatures in the tropics cause higher rates of metabolism, ecological dynamics and coevolutionary processes, which generate and maintain higher biodiversity.

Q.49) Ans: c

Exp:

- **Flagship Species** : The species selected to act as an ambassador, icon or symbol for a defined habitat, issue, campaign or environmental cause and by focussing on them, their habitat and other species which share the same habitat can also be conserved
- **Iconic Species** : The species having cultural significance and are involved in traditional activities and are broadly recognised for their existence and aesthetic values
- **Guilds** : A group of species with similar ecological features, similar requirements and play a similar role within a community
- **Umbrella Species** : The species requiring large areas of natural habitat for their continued survival

Q.50) Ans: a

Exp:



- **Statement 1 is Correct** : **Keystone Species** is a species whose addition to or loss from an ecosystem leads to major changes in the occurrence of at least one other species. All top Predators are considered as keystone Species because they regulate all other animals population Indirectly
- **Statement 2 is Incorrect** : Ecosystem Engineers are like Foundation species. Ecosystem engineers modify, create, and maintain habitats. They modify their habitats through their own biology or by physically changing biotic and abiotic factors in the environment.
- **Statement 3 is Incorrect** : Keystone species have low functional redundancy. This means that if the species were to disappear from the ecosystem, no other species would be able to fill its ecological niche. The ecosystem would be forced to radically change, allowing new and possibly invasive species to populate the habitat.

Q.51) Ans: b

Exp:

- **Statement 1 is Incorrect** : Foundation species is a dominant primary producer both in terms of abundance and Influence. ex- kelps in Kelp forest and Corals in coral reefs
- **Statement 2 is Correct** : In Ecology, the term foundation species is used to refer to a species that has a strong role in restructuring a Community. The activities of foundation species physically modify the environment and produce and maintain habitats that benefit other organisms using that benefit. The presence of foundation species has the ability to either reduce or increase species diversity depending on its particular role in a specific ecosystem.
 - For example -: Corals are very unique species and one of the most thought-of species for this type of example. Corals produce reef structures in marine ecosystems that provide a huge range of species with habitats, protection against predation and breeding grounds. Without Corals, these type of marine ecosystems would not be sustainable
- **Statement 3 is Correct** : A foundation species can occupy any trophic level in a food web. It can be primary producer, herbivores or Predators

Q.52) Ans: b

Exp:

- **Realms** are large spatial regions within which ecosystems share a broadly similar biota. A realm is a continent or subcontinent sized area with unifying features of geography and fauna & flora. The Indian region is composed of two realms.
 - They are:
 - the Himalayan region represented by Palearctic Realm and
 - the rest of the sub-continent represented by Malayan Realm
- In the world, Eight terrestrial biogeographic realms are typically recognised. They are
 - Nearctic Realm
 - Palaearctic Realm
 - Afrotropical Realm
 - Indomalayan Realm
 - Oceania Realm
 - Australian Realm
 - Antarctic Realm
 - Neotropical Realm

Q.53) Ans: b

Exp:

- **Statement 1 is Incorrect** : The Ten Biogeographic zone as a percentage of total Geographical area are as
 - Andaman & Nicobar Island : 0.8% (of Total Geographical Area)
 - Coastal Region : 2.5%
 - North East region : 5.2%
 - Gangetic Plains : 10.8%
 - Deccan Plateau : 42%
 - Western Ghats : 4%
 - Semi Arid Region : 16.6%
 - Indian Desert Zone : 6.6 %
 - Himalayan Zone : 6.4 %
 - Trans-Himalayan Zone : 5.6%
- **Statement 2 is Correct** : Andaman and Nicobar Islands constitute 0.3 per cent of the total geographical area and are one of the three tropical moist evergreen forests zones in India. The islands house an array of flora and fauna not found elsewhere. These islands are centres of high endemism and contain some of India's finest evergreen forests and support a wide diversity of corals. In India, endemic island biodiversity is found only in the Andaman and Nicobar Islands. Some of the endemic fauna of Andaman & Nicobar islands include Narcondam hornbill, South Andaman krait etc.



- **Statement 3 is Correct** : Rogers and Panwar of the Wildlife Institute of India (WII) outlined a scheme to divide India zoogeographically in 1986 while planning a protected area network for India.[13] The scheme divided India into 10 biogeographic zones, and each zone is further subdivided into biogeographic provinces, which total 27 in number. 1. Trans Himalayan zone. 2. Himalayan zone 3. Desert zone. 4. Semi Arid zone. 5. Western ghat zone. 6. Deccan plateau zone. 7. Gangetic plain zone. 8. North east zone. 9. Coastal zone. 10. Islands

Q.54) Ans: a

Exp:

- **Statement 1 is Correct** : Acclimatization is the process in which an individual organism adjusts to a change in its environment (such as a change in altitude, temperature, humidity, photoperiod, or pH), allowing it to maintain performance across a range of environmental conditions.
- **Statement 2 is Incorrect** : Acclimatization occurs in a short period of time (hours to weeks), and within the organism's lifetime (compared to adaptation, which is a development that takes place over many generations).
- **For example** : Humans who migrate permanently to a higher altitude naturally acclimatize to their new environment by developing an increase in the number of red blood cells to increase the oxygen carrying capacity of the blood, in order to compensate for lower levels of oxygen intake

Q.55) Ans: C

Exp:

In the last 500 years, human activity is known to have forced more than 800 species to extinction.

- Some examples of **recent extinctions** include the dodo (Mauritius), quagga (Africa), Himalayan Quail, Thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger.
- The **sloth bear** is a myrmecophagous bear species native to the Indian subcontinent. It is listed as Vulnerable on the IUCN Red List, mainly because of habitat loss and degradation
- **Species extinct from India**
- **Flora**
 - As per information given by the BSI, 18 species of plants — four non-flowering and 14 flowering — have gone extinct.

- The notable among them are Lastreopsis wattii, a fern in Manipur discovered by George Watt in 1882 and three species from the genus Ophiorrhiza (Ophiorrhiza brunonis, Ophiorrhiza caudate and Ophiorrhiza radican), all discovered from peninsular India.
- Corypha taliera Roxb, a palm species discovered in Myanmar and the Bengal region by William Roxburgh is also extinct.

• Fauna

- Cheetah (Acinonyx jubatus)
- Sumatran rhinoceros (Dicerorhinus sumatrensis) are considered extinct in India.
- The pink-headed duck (Rhodonessa caryophyllacea) is feared extinct since 1950
- The Himalayan quail (Ophrysia superciliosa) was last reported in 1876.
- Some General cause for extinction of species
 - **Climate Change** : Climate change could destroy terrestrial organisms. During the end of the last ice-age, most of the megafauna were unable to adapt to the changing warm temperatures. They died due to lack of food and hunting by early humans
 - **Disease** : The Black Death wiped out one-third of the European population in the Middle Ages.
 - **Loss of habitat** : Every animal has its own comfort zone where it can breed and raise its young ones. For eg., a bird is comfortable only on the branch of a tree. Due to the expansion of human civilization and industrialization, the forests have been destroyed which are an abode to most of the animals. Due to lack of space and eventually food, the populations of many organisms have been minimised.
 - **Lack of Genetic Diversity** : Once the number of species starts decreasing, the gene pool of that species grows smaller. Eventually, there is a lack of genetic diversity. For eg., due to habitat loss, the African cheetahs have a considerably low genetic diversity.
 - **Better Adapted Competition** : The better-adapted populations win over the ones that lag behind. For eg., the pre-historic

mammals were better adapted than the dinosaurs. The ones which are well-adapted survive, while the others become extinct

- **Pollution** : The pollution from the industries and vehicles have led to a drastic change in the oxygen levels in the atmosphere as well as water. This has led to the extinction of most of the aquatic as well as terrestrial species

Q.56) Ans: d

Exp:

- **Statement 1 is Incorrect** : A **Hope Spot** is an area of an ocean that needs special protection because of its wildlife and significant underwater habitats.
 - Less than 1% of the oceans are protected. By identifying such hope spots, IUCN wants to "scale up the marine protection necessary for sustainable development of the ocean".
- **Statement 2 is Incorrect** : Cool-spots' are the world's last refuges where high numbers of threatened species still persist. Cool-spots could be the result of protection or because of intact habitat that has not been cleared yet.
- **Statement 3 is Incorrect**
 - Andaman and Nicobar Islands and Lakshadweep islands have recently been named as the new "hope spots" by the International Union for Conservation of Nature (IUCN) and Mission Blue, an organization involved in the study of oceans
 - Some of the "cool spots" identified include parts of the Amazon rainforest, Andes Mountains, and tundra and boreal forests of Russia and North America

Q.57) Ans: b

Exp:

- **Statement 1 is correct** : A National park or Sanctuary is declared by state government via notification if it consider such area (within or outside any reserve forest) of adequate ecological, faunal, floral, geomorphological, natural or zoological significance
 - Central Govt. can also declare a national park or wildlife sanctuary via notification if it considers such an area (within or outside any reserve forest) of adequate

ecological, faunal, floral, geomorphological, natural or zoological significance.

- **Statement 2 is Incorrect** : Boundaries of national parks and sanctuaries shall be fixed by a notification. However no alterations can be made except on recommendation of the National Board of Wildlife
- **Statement 3 is Correct** : National board of wildlife shall make recommendations on setting up and management of protected areas.
 - State Wildlife Boards shall advise state governments on selection and administration of such areas

Q.58) Ans: a

Exp:

- The Wildlife (Protection) Act, 1972 provides for the establishment of Protected Areas in India. National parks and wildlife sanctuaries are protected natural habitats, declared by the government of a country according to the regulations from the IUCN (The World Conservation Union) to preserve the wildlife through the conservation of ecosystems
- **Statement 1 is Correct** : National parks enjoy a higher degree of protection as certain activities like grazing, which is regulated in sanctuaries, isn't permitted in national parks. Only certain areas can be visited and only activities permitted by the chief wildlife warden of the state are allowed in the park.
- **Statement 2 is Correct** : Sanctuaries are formed for particular species but not national parks which may be for multiple species.
 - Wildlife Sanctuaries are famous for the conservation of wildlife, which includes animals, insects, microorganisms, birds, etc. of different genes and species. On the other hand, National Parks are highly known for preserving the flora, fauna, landscape and historical objects.
- **Statement 3 is Incorrect** : National Park cannot be downgraded to the status of a 'sanctuary'. The International Union of Conservation of Nature, IUCN, has grouped wildlife sanctuaries in Category IV of protected areas. Hence National park, since accorded a higher degree of Protection, can't be downgraded.

Q.59) Ans: b



Exp:

- **Statement 1 is Incorrect & 2 is Correct :**
Community Reserve is declared by the State Government in any private or community land.. The land should not be within a National Park, Sanctuary or a Conservation Reserve.
- **Statement 3 is Correct :** Rights of people living inside a Community Reserve are not affected. It is basically an area where an individual or a community has volunteered to conserve wildlife and its habitat

Q. 60) Ans: b

Exp:

- **Statement 1 is correct :** The State Government declares a conservation reserve in an area owned by the Central or State government. The aim of conservation reserves is to protect landscapes, seascapes, flora and fauna and their habitat
 - **Statement 2 is Incorrect :** Rights of people are not affected in a conservation reserve
 - **Statement 3 is Correct :** Tiruppadaimarathur conservation reserve near Thirunelveli District of Tamil Nadu, declared in 2005, is the first Conservation Reserve in India
 - **Statement 4 is Incorrect :** Developmental activities can be taken up in Conservation reserves but with the approval of the State wildlife board. Agricultural activities can also be taken up with permission from the state wildlife board.
- **Statement 2 is Incorrect :** The boundaries of Tiger reserves cannot be altered without recommendation of National board for Wild Life and without advice of Tiger Conservation Authority
 - **Statement 3 is Correct :** National Tiger Conservation Authority (NTCA) is a statutory body under the Ministry of Environment, Forests and Climate Change.
 - It was established in 2005 following the recommendations of the Tiger Task Force.
 - It was constituted under enabling provisions of the Wildlife (Protection) Act, 1972, as amended in 2006, for strengthening tiger conservation, as per powers and functions assigned to it.
 - **Statement 4 is correct :** St. Petersburg Declaration on Tiger Conservation
 - This resolution was adopted In November 2010, by the leaders of 13 tiger range countries (TRCs) assembled at an International Tiger Forum in St. Petersburg, Russia
 - It aimed at promoting a global system to protect the natural habitat of tigers and raise awareness among people on white tiger conservation.

Q.62) Ans: d

Exp:

- **Statement 1 is Correct :** Biosphere reserves or natural reserves are multipurpose protected areas with boundaries circumscribed by legislation. Biosphere reserves or natural reserves are multipurpose protected areas with boundaries circumscribed by legislation. The concept of Biosphere Reserves was launched in 1975 as a part of UNESCO's 'Man and Biosphere Programme, dealing with the conservation of ecosystems and the genetic resources contained therein.
 - **Statement 2 is correct :** The buffer zone, adjoins or surrounds core zone. It is Used for scientific research, monitoring, training and education. The activities are managed in this area in the ways that help in protection of the core zone in its natural environment.
 - **Statement 3 is Correct :** Biosphere Reserve demonstrates integrated management of land, water and Biodiversity. Limited economic activity(sand and stone mining are permitted)
- A National Park or Wildlife Sanctuary that is considered significant for protecting tigers can be additionally designated as a Tiger Reserve.
 - There are 50 tiger reserves in India governed by Project Tiger which is administered by the National Tiger Conservation Authority (NTCA)
 - **Statement 1 is correct :** Tiger reserves are declared by NTCA via Wild Life (Protection) Amendment Act, 2006.
 - If the state government wishes to declare an area as Tiger reserve , they can forward their proposal though NTCA. The Central govt. NTCA can also advise the state government to forward the proposal.

Q.61) Ans: b

Exp:



Q.63) Ans: d

Exp:

- **Statement 1 is Correct** : India has a history of religious/cultural traditions that emphasised protection of nature.
 - In many cultures, tracts of forest were set aside, and all the trees and wildlife within were venerated and given total protection.
 - Such sacred groves are found in Khasi and Jaintia Hills in Meghalaya, Aravalli Hills of Rajasthan, Western
 - Ghat regions of Karnataka and Maharashtra and the Sarguja, Chanda and Bastar areas of Madhya Pradesh.
- **Statement 2 is Correct** : Botanical garden refers to the scientifically planned collection of living trees, shrubs, herbs, climbers and other plants from various parts of the globe.
- Purpose of botanical gardens:
 - To study the taxonomy as well as the growth of plants.
 - To study the introduction and acclimatisation process of exotic plants.
 - It acts as a germplasm collection.
 - It helps the development of new hybrids.
 - It augments conserving rare and threatened species.
 - It facilitates training of staff.
 - It acts as a source of recreation.
- **Statement 3 is Correct** : Zoo is an establishment, whether stationary or mobile, where captive animals are kept for exhibition to the public and includes a circus and rescue centers but does not include an establishment of a licensed dealer in captive animals.
 - The initial purpose of zoos was entertainment, over the decades zoos have got transformed into centers for wildlife conservation and environmental education

Q.64) Ans: d

Exp:

- **Statement 1 is Correct** : The Wildlife (Protection) Act, 1972 was passed by the Parliament under Article 252 of the Constitution, it was intended to provide a comprehensive National framework for wildlife protection and to adopt a conservation strategy for specified endangered species and

provide for protection of all species in specified areas

- **Statement 2 is Incorrect** : Schedule 6 states the list of specified endemic plants which are prohibited for cultivation and planting
- **Statement 3 is Incorrect** : Hunting of Wild Animals to be permitted in certain cases (Section 11)
 - The Chief Wildlife Warden may, if he/she is satisfied that any wild animal specified in Sch. 1 has become dangerous to human life or is so disabled or diseased as to be beyond recovery, by order in writing and stating the reasons therefore, permit any person to hunt such animal or cause animal to be hunted; Provided that no wild animal shall be ordered to be killed unless the Chief Wild Life Warden is satisfied that such animal cannot be captured, tranquilized or translocate
- **Statement 4 is Correct** : Management of zoos is also under ambit of this act to provide shelter and protect the animals which are not in danger but need protection and security

Q.65) Ans: b

Exp:

The claims by different political parties over significant reduction in poaching of Greater One-Horned Rhino has become an issue in Assam Assembly Elections.

- **Statement 1 is Incorrect** : There are three species of rhino in Asia—Greater one-horned (Rhinoceros unicornis), Javan and Sumatran. Only the Great One-Horned Rhino is found in India. Javan and Sumatran Rhino are critically endangered and the Greater one-horned (or Indian) rhino is vulnerable in IUCN Red List. All three listed under Appendix I (CITES). Greater one-horned rhino is listed under the Schedule I of the Wildlife Protection Act, 1972.
- **Statement 2 is Correct** : The five rhino range nations (India, Bhutan, Nepal, Indonesia and Malaysia) have signed a declaration 'The New Delhi Declaration on Asian Rhinos 2019' for the conservation and protection of the species
- **Statement 3 is Correct** : The Indian Rhino is grey-brown in colour and has one horn on its snout. This one horn is its most distinguishing feature. Aside from its eyelashes, ear fringes and tail brush it is largely hairless.

- Its single horn is found in males and females although not in newborn calves. The horn is made of keratin, similar to what's found in human fingernails. The horns started to show after six years. The length of the horn is about 25 cm but horns upto 36 cm have also been measured.
- **Statement 4 is correct** : The Indian Rhino is grey-brown in colour and has one horn on its snout. This one horn is its most distinguishing feature. Aside from its eyelashes, ear fringes and tail brush it is largely hairless
 - Its single horn is found in males and females although not in newborn calves. The horn is made of keratin, similar to what's found in human fingernails. The horns started to show after six years. The length of the horn is about 25 cm but horns upto 36 cm have also

Q.66) Ans: D

Exp.

- Dimethylsulfide (DMS) is a naturally-produced gas that has been called the 'anti-greenhouse gas' due to its potential to cause cloud formation, which alters how much solar radiation makes it through the atmosphere. In sunlit zones of the ocean, microbial activity can produce enough DMS that the ocean serves as a net source to the atmosphere, especially in sub-Arctic regions. It has been observed that warmer waters will lead to more DMS production in high-latitude waters.
- DMS originates primarily from DMSP, a major secondary metabolite in some marine algae. DMS is the **most abundant biological sulfur compound emitted to the atmosphere**. Emission occurs over the oceans by phytoplankton. DMS is also produced naturally by bacterial transformation of dimethyl sulfoxide (DMSO) waste that is disposed of into sewers, where it can cause environmental odor problems.

Q.67) Ans: A

Exp.

- **Statement 1 is correct** : To get a complete picture of Earth's temperature, scientists combine measurements from the air above land and the ocean surface collected by ships, buoys and sometimes satellites, too. The temperature at each land and ocean station is compared daily to

what is 'normal' for that location and time, typically the long-term average over a 30-year period. The differences are called an 'anomalies' and they help scientists evaluate how temperature is changing over time

- **Statement 2 is incorrect** : Worldwide, 2006-2015 was the warmest decade on record since thermometer-based observations began nearly 150 years ago.

Q.68) Ans: D

Exp.

- Scientists use four major datasets to study global temperature. The UK Met Office Hadley Centre and the University of East Anglia's Climatic Research Unit jointly produce HadCRUT4.

Extending back to 1850 and frequently updated, HadCRUT4 is the longest data set of its type. HadCRUT4 is a combination of the global land surface temperature data set, CRUTEM4 and the global SST data set, HadSST3. HadCRUT4 is different from the most closely comparable products (e.g. NASA GISTEMP and NOAA MLOST) in that no interpolation is performed. As such, there are many coverage gaps; this may be a limiting factor in regional analyses but hemispheric and global means are very consistent with other data sets.

Q.69) Ans: A

Exp.

Global Warming Potential (GWP) is the ability of a greenhouse gas to absorb heat compared to carbon dioxide over a specified period of time, from 20 to 500 years. The time frame is important because each gas has a different rate at which it is removed from the atmosphere. For each time period, carbon dioxide is always set at "1", and other greenhouse gases are compared to carbon dioxide for the same timeframe. For example, the sulfur hexafluoride's GWP at 20 years is 15,100, meaning it has 15,100 times more warming potential than carbon dioxide in that timeframe.

Q.70) Ans: B

Exp.

- Mesoscale meteorology is the study of atmospheric phenomena with typical spatial scales between 10 and 1000 km. Examples of mesoscale phenomena include thunderstorms, gap winds, downslope windstorms, land-sea breezes, and squall lines. Many of the weather phenomena that most directly impact human activity occur on the mesoscale. Research in mesoscale meteorology has been spurred by recent advances in observational and numerical modeling capabilities that have greatly improved the tools used by atmospheric scientists to study mesoscale weather systems

– “bottom-up” and “top-down” approaches, which primarily have been used to assess the economic potential

- Bottom-up studies are based on assessment of mitigation options, emphasizing specific technologies and regulations. They are typically sectoral studies taking the macro-economy as unchanged. Sector estimates have been aggregated, as in the TAR, to provide an estimate of global mitigation potential for this assessment
- Top-down studies assess the economy-wide potential of mitigation options. They use globally consistent frameworks and aggregated information about mitigation options and capture macro- economic and market feedbacks.

Q.71) Ans: C

Exp.

- **Statement 1 is correct:** The objective of the Coupled Model Intercomparison Project (CMIP) is to better understand past, present and future climate changes arising from natural, unforced variability or in response to changes in radiative forcing in a multi-model context. This understanding includes assessments of model performance during the historical period and quantifications of the causes of the spread in future projections.
- **Statement 2 is correct:** The European Space Agency has been selected by the World Climate Research Programme (WCRP) as the host for a new Coupled Model Intercomparison Project (CMIP) International Project Office (IPO), which should be fully operational from September 2021.

Q.73) Ans: C

Exp:

- The term decoupling refers to breaking the link between “environmental bads” and “economic goods.” Decoupling environmental pressures from economic growth is one of the main objectives of the OECD Environmental Strategy for the First Decade of the 21st Century, adopted by OECD Environment Ministers in the starting of the millennium.

Decoupling occurs when the growth rate of an environmental pressure is less than that of its economic driving force (e.g. GDP) over a given period. Decoupling can be either absolute or relative.

- Absolute decoupling is said to occur when the environmentally relevant variable is stable or decreasing while the economic driving force is growing.
- Decoupling is said to be relative when the growth rate of the environmentally relevant variable is positive, but less than the growth rate of the economic variable.

Q.72) Ans: A

Exp.

- The concept of “mitigation potential” has been developed to assess the scale of GHG reductions that could be made, relative to emission baselines, for a given level of carbon price (expressed in cost per unit of carbon dioxide equivalent emissions avoided or reduced). Mitigation potential is further differentiated in terms of “market potential” and “economic potential”.

Mitigation potential is estimated using different types of approaches. There are two broad classes

Q.74) Ans: C

Exp.

- **Statement 1 is correct:** The Climate Change Performance Index published by Germanwatch,

CAN International and the NewClimate Institute evaluates and compares the climate protection performance of 57 countries and the EU.

- **Statement 2 is incorrect:** It has been published annually since 2005 and is an independent monitoring tool to track climate change mitigation practices of 57 countries and the EU.
- **Statement 3 is correct:** India is ranked at 10th place in the 'Climate Change Performance Index (CCPI) 2021', released on 7th December 2020, with a score of 63.98. It is for the second time in a row that India emerged among the top 10 countries with higher climate performance from among 58 economies.

Q.75) Ans: B

Exp.

- The World Meteorological Organization issued the first state of the climate report in 1993. The report was initiated due to the concerns raised at that time about projected climate change. Major findings of the report are:-
 - In 2020, the Global Mean surface Temperature (GMST) was $1.2 \pm 0.1^\circ\text{C}$ warmer than the pre-industrial baseline (1850-1900). Despite developing La Niña cooling conditions, 2020 was one of the three warmest years on record as per the report.
 - Emission and Concentrations of the major greenhouse gases in the air increased during the 2019- 2020 phase.
 - Heating of the oceans (melting of glaciers) and rising of the sea levels (La Niña induced cooling) occurring simultaneously.
 - The most acute impacts of weather and climate are felt during extreme meteorological events such as heavy rain and snow, droughts, heatwaves, cold waves, and storms, including tropical storms.

Q.76) Ans : D

Exp :

About Sree Narayana Guru

- Sree Narayana Guru was born on August 22, 1856 in the pretty hamlet of Chempazhanthi near Thiruvananthapuram. His father was Madan Asan, a farmer, and mother Kutti Amma and he had three sisters. He was named by his father as Narayanan; he affectionately called him Nanu, a short name for Narayanan.

Works of Sree Naryana Guru

Advaitha Deepika, Anukamba Dasakam, Arivu, Athma Vilasam, Atmopadesa Āśatakam, Bhadrakaalyashtakam, Chijanda Chinthakam, Daiva Chinthanam - 1 & 2, Daiva Dasakam, Jathi Lakshanam, Jathi Nirnayam, Jeevakarunya Panchakam, Shiva Shathakam, Swanubavageethi.

- Right from his childhood Nanu showed great interest in the stories narrated to him by his father from the great epic Ramayana and Mahabharata. After this elementary education he became the disciple of Raman Pillai Asan, a great Sanskrit scholar of Puthuppally Varanappally family in Central Travancore at the age of 21.
- He started Sree Narayana Guru Dharma Paripalana (SNDP) Movement among the Ezhavas of Kerala, who were a backward caste of toddy-tappers and were considered to be untouchables, denied education and entry into temples.
- His philosophy always advocated social equality, education for all, and spiritual enlightenment. In 1903, he established the Sree Narayana Dharma Paripalana Yogam (SNDP).
- Narayana Guru, himself from the Ezhava caste, took a stone from the Neyyar river and installed it as a Sivalinga at Aruvippuram in 1888. It was intended to show that consecration of an idol was not the monopoly of the higher castes. With this he began a revolution that soon led to the removal of many discriminations in Kerala's society.
- The movement (Aruvippuram movement) drew the famous poet Kumaran Asan as a disciple of Narayana Guru. In 1889, the Aruvippuram Kshetra Yogam was formed which was decided to expand into a big organisation to help the Ezhavas to progress materially as well as spiritually.

On the Aug 23 the birth anniversary of Sree Narayana Guru was celebrated in Kerala.

Q.77) Ans: B

Exp:

Changpa Tribe

- The **Changpa** are a semi-nomadic Tibetan people found mainly in the Changtang in Ladakh. The Changpas are **nomadic pastoralists** who predominantly rear sheep and goats.
- The Changpa nomads have been shepherding their world-class **pashmina goats** for centuries and these goats are one of their main sources of income. The famous pashmina wool is only obtained from a special breed of goats known as *ChangRa*, meaning the goat of the Changpas. These *ChangRa* are grazed on cold and high lands to produce fine pashmina wool.
- The Changpa nomads are **Buddhists**. One of the main purifying rituals of the Changpas is known as *Jyabten* where they make different kinds of food as an offering to Lord Padmasambhava. *Tangpe Chona* is a festival of Changpas.

Q.78) Ans: C

Exp:

- **Statement 1 is correct:** **Yuktdhara** is a new geospatial planning portal. It will help in facilitating new MGNREGA assets using remote sensing and geographic information system-based data. The portal will serve as a repository of assets (geotags) created under various national rural development programmes i.e. MGNREGA, Integrated Watershed Management Programme, Per Drop More Crop programme and Rashtriya Krishi Vikas Yojana, along with field photographs.
- **Statement 2 is correct:** The "Yuktdhara" portal was launched by Ministry of Rural Development and Panchayati Raj. Yuktdhara based plans will be prepared by **grassroot functionary** and verified by appropriate authorities for relevance and resource allocation. This would ensure quality of plan and enable a long-term monitoring of the assets created over the years.

Q.79) Ans : C

Exp :

The Hazaras

- The **Hazaras** are a Persian-speaking ethnic group native to, and primarily residing in the Hazarajat region in central Afghanistan and generally scattered throughout Afghanistan.
- They are one of the third largest ethnic groups in Afghanistan, and are also a significant minority group in neighbouring Pakistan
- Babur, founder of the Mughal Empire in the early 16th century, records the name *Hazara* in his autobiography. He referred to the populace of a region called Hazaristan, located west of the Kabulistan region. Hazaras are considered to be one of the most oppressed groups in Afghanistan.

Q.80) Ans : D

Exp :

Both Statements are correct

- The Prime Minister's **Shram Awards** were instituted in 1975 by the Government of India This



The infographic details the categories of the Prime Minister Shram Awards:

- SHRAM RATNA**: This is the highest Award (one in number) among the Shram Awards and carries a cash award of Rs. 2,00,000/- along with a 'Sanad'.
- SHRAM BHUSHAN**: Total number of Shram Bhushan Award is four. It carries a cash award of Rs. 1,00,000/- and a 'Sanad'.
- SHRAM VIR/ SHRAM VEERANGANA**: Total number of Shram Vir/Shram Veerangana Awards is twelve. It carries a cash award of Rs. 60,000/- and a 'Sanad'.
- SHRAM SHREE/ SHRAM DEVI**: Total number of Shram Shree/Shram Devi Awards is sixteen. It carries a cash award of Rs. 40,000/- and a 'Sanad'.

The awards are conferred by the Ministry of Labour & Employment, Government of India. Social media handles: @LabourMinistry, /LabourMinistry, @MinistryofLabour&Employment, @LabourMinistry.

national award is conferred on workers for outstanding contributions that improve productivity, innovation, and indigenization, resulting in saving foreign exchange.

- The Awards will be presented to the workmen as defined in the Industrial Disputes Act, 1947 and employed in the Departmental Undertakings of the Central and State Governments, Central and State Public Sector Undertakings as also private sector and having minimum of 500 employees on roll and engaged in manufacturing and productive processes and whose performance is assessable.
- Workmen solely engaged in routine service duties would not be eligible.

Q.81) Ans : A

Exp :



About the SUJALAM

- The Ministry of Jal Shakti began 'SUJALAM', a '100 days campaign' as part of the 'Azadi Ka Amrit Mahotsav' celebrations to create more and more ODF Plus villages by undertaking waste water management at village level particularly through creation of 1 million Soak-pits and also other Grey water management activities.
- The effort of campaign would be directed towards achieving the ODF plus status for villages across the country in an accelerated manner in a short time. The Campaign has started from today i.e. 25th August, 2021 and will continue to run for the next 100 days.
- The campaign will not only build desired infrastructure i.e. soak pit for management of greywater in villages but will also aid in sustainable management of waterbodies. The disposal of waste water and clogging of waterbodies in the villages or on the outskirts of the villages remain one of the major problems. The Campaign would help in management of the wastewater and in turn would help to revive the waterbodies.

Q.82) Ans : C

Exp :

Above 3 places are among the 4 new Ramsar sites declared under the Ramsar Convention.

- **Haryana's Bhindawas Wildlife Sanctuary** is a human-made freshwater wetland. It is also the largest in Haryana. Over 250 bird species use the sanctuary throughout the year as a resting and roosting site. The site supports more than 10 globally threatened species including the endangered Egyptian Vulture, Steppe Eagle, Pallas's Fish Eagle, and Black-bellied Tern.
- **The Sultanpur National Park in Haryana** supports more than 220 species of resident, winter migratory and local migratory waterbirds at critical stages of their life cycles. More than 10 of these are globally threatened, including the critically endangered sociable lapwing, and the

- The Ramsar Convention is an international treaty for the conservation and wise use of wetlands. It is named after the Iranian city of Ramsar, on the Caspian Sea, where the treaty was signed on February 2, 1971.
- The 46 Ramsar sites in India include the Chilika Lake in Odisha, Keoladeo National Park in Rajasthan, Harike Lake in Punjab, Loktak Lake in Manipur and Wular Lake in Jammu and Kashmir.

endangered Egyptian Vulture, Saker Falcon, Pallas's Fish Eagle and Black-bellied Tern.

- **The Thol Lake Wildlife Sanctuary in Gujarat** lies on the Central Asian Flyway and more than 320 bird species can be found here. The wetland supports more than 30 threatened waterbird species, such as the critically endangered White-rumped Vulture and Sociable Lapwing, and the vulnerable Sarus Crane, Common Pochard and Lesser White-fronted Goose.
- **The Wadhvana Wetland in Gujarat** is internationally important for its birdlife as it provides wintering ground to migratory waterbirds, including over 80 species that migrate on the Central Asian Flyway. They include some threatened or near-threatened species such as the endangered Pallas's fish-Eagle, the vulnerable Common Pochard, and the near-threatened Dalmatian Pelican, Grey-headed Fish-eagle and Ferruginous Duck.

Q.83) Ans : C

Exp :

- Statement 1 is NOT CORRECT. Sovereign Gold Bond Scheme will help reduce India's over dependence on gold imports but not drastically.
- Statement 2 is NOT CORRECT. Holder of the bond will earn fixed rate of 2.5% per annum, payable semi-annually. The interest on Gold Bonds shall be taxable as per the provision of the Income Tax Act, 1961.



- Statement 3 is CORRECT. Bonds can be used as collateral for loans. The loan-to-value (LTV) ratio is to be set equal to ordinary gold loan mandated by the Reserve Bank from time to time.
- The bonds will be restricted for sale to resident Indian entities, including individuals, HUFs, trusts, universities and charitable institutions.
- Denomination and tenor: The bonds will be denominated in multiples of gram(s) of gold with a basic unit of 1 gram. The tenor will be for a period of 8 years with exit option from the 5th year to be exercised on the interest payment dates.
- Minimum and Maximum limit: The minimum permissible investment limit will be 1 gram of gold, while the maximum limit will be 4 kg for individual, 4 kg for Hindu Undivided Family and 20 kg for trusts and similar entities per fiscal (April-March) notified by the government from time to time.

Q.84) Ans: D

Exp:

- The Chakmas and Hajongs, originally residents of the Chittagong Hill Tracts of former East Pakistan, had to flee when their land was submerged by the Kaptai dam project in the 1960s.
- The migrants were settled with a rehabilitation plan and allotted permanent land and provided with financial assistance depending on the size of their families to help rebuild their lives. As per the 2011 census, there are 47,471 Chakmas and Hajongs in Arunachal Pradesh.

Q. 85) Ans: A

Exp

- Quit India movement and Rise of future leaders: This movement also gave some future prominent leaders such as Biku Patnaik, Aruna Asif Ali, Ram Manohar Lohia, Sucheta Kriplani, J.P. Narayan, etc. These leaders were helping the movement through underground activities.

Q.86) Ans: C

Exp:

- Recently, West Africa's first case of the Marburg virus was confirmed in Guinea.
- Marburg virus disease is a highly virulent disease that causes hemorrhagic fever.
- It is in the same family as the virus that causes Ebola virus disease.
- Marburg is transmitted to people from fruit bats. The virus spreads among humans through direct contact with the bodily fluids of the patient, surfaces and materials

Q.87) Ans: C

Exp:

Option C is correct

About SAMRIDH Scheme

- SAMRIDH stands for Start-up Accelerators of MeitY for pRодукt Innovation, Development and growth. It was recently launched by the Ministry of Electronics & Information Technology.
- It aims to create a conducive platform for Indian Software Product start-ups to enhance their products and secure investments for scaling their business.
- Firstly, the scheme will focus on accelerating the 300 start-ups by providing customer connect, investor connect, and international immersion in the next three years.
- Secondly, an investment of up to ₹ 40 lakh to the start-up based on the current valuation and growth stage of the Start-Up will be provided through selected accelerators.
- The scheme will be implemented by MeitY Start-up Hub.

Q.88) Ans: D

Exp:

Option D is correct

- About Kaziranga National Park
- Kaziranga National Park lies partly in Golaghat District and partly in Nagaon District of Assam.
- It is the oldest park in Assam covering an area of 430 sq km along the river Brahmaputra on the North and the Karbi Anglong hills on the South.



- National Highway 37 passes through the park area.
- It is a world heritage site that is famous for the Great Indian one-horned rhinoceros.
- It was declared a National Park in 1974.
- It is recognized as an Important Bird Area by BirdLife International for the conservation of avifaunal species.
- The landscape of Kaziranga is of sheer forest, tall elephant grass, rugged reeds, marshes & shallow pools.
- It is inhabited by the world's largest population of one-horned rhinoceroses, as well as many mammals, including tigers, elephants, panthers and bears, and thousands of birds.
- Recently, Kaziranga National Park (KNP) has become the first national park in India to be equipped with satellite phones.

Extra Edge by OnlyIAS

- About Manas National Park:
- Manas National Park is a Project Tiger reserve, an elephant reserve and a biosphere reserve. It is a UNESCO world heritage site under the natural site category. It is located in the Himalayan foothills in Assam on the northern side of the Brahmaputra river. It is contiguous with the Royal Manas National Park in Bhutan.
- About Nameri National Park:
- Nameri National Park is located in the foothills of the Eastern Himalayas in Assam. The Pakhui (Pakke) Sanctuary of Arunachal Pradesh adjoins the Park on its North-Eastern point. It consists of semi-evergreen, moist deciduous forests with cane and bamboo brakes and narrow strips of open grassland along rivers.
- About Raimona National Park:
- The Raimona National Park is the 6th national park in Assam and is located within the Bodoland Territorial Region. The area of the park includes the northern part of the notified Ripu Reserve Forest, which forms the western-most buffer to the Manas National Park that straddles the India-Bhutan border. It is bounded on the

west by the Sonkosh river and the Saralbhanga river on the east. It is famous for Golden Langur, an endemic species which has been named as the mascot of Bodoland region.

Q.89) Ans: B

Exp:

Option B is correct:

About PM-DAKSH Scheme:

- Pradhan Mantri Dakshta Aur Kushalta Sampann Hitgrahi (PM-DAKSH) Yojana is being implemented by the Ministry of Social Justice and Empowerment from 2020-21.
- It is a National Action Plan for the marginalized persons of SC, OBC, Economically backward classes (EBC), Senior Citizens, De-notified Tribes (DNTs), Safai karamcharies including waste pickers, and women.
- Through this scheme, MoSJ&E empowers the socially, educationally and economically marginalized sections of the society, by providing skill development training programmes to the eligible target groups on,
 - o Up-skilling/Re-skilling,
 - o Short Term and Long-Term Training Programmes, and
 - o Entrepreneurship Development Program (EDP).
- These training programs are being implemented through Government Training Institutes, Sector Skill Councils constituted by the Ministry of Skill Development and Entrepreneurship and other credible institutions.
- The following Apex Corporations functioning under the MoSJ&E provide loans at concessional interest rates and skill development training to the above target groups for self-employment,
 - o National Scheduled Castes Finance and Development Corporation,
 - o National Backward Classes Finance and Development Corporation,
 - o National Safai Karamchari Finance and Development Corporation

Q.90) Ans: B

Exp:

Option B is correct

About National Mission on Edible Oil-Oil Palm



- It is a new centrally sponsored scheme and will subsume the current National Food Security Mission-Oil Palm programme.
- Under this scheme, it is proposed to cover an additional area of 6.5 lakh hectare (ha.) (6.5 million hectare would be equal to 65 lakh hectare) for oil palm till the year 2025-26 and thereby reaching the target of 10 lakh hectares ultimately.
- It will have a special focus on the Northeast region and the Andaman and Nicobar Islands.
- Price assurance to the oil palm farmers on the lines of the minimum support price (MSP) for the Fresh Fruit Bunches (FFBs) from which oil is extracted by the industry. This will be known as the Viability Price (VP).
- The assurance to the farmers will be in the form of the viability gap funding and the industry will be mandated to pay 14.3% of the CPO (Crude Palm Oil) price which will eventually go up to 15.3%.
- To give impetus to the North-East and Andaman, the Government will additionally bear a cost of 2% of the CPO price.
- Assistance to farmers for planting materials of Rs 29,000 per hectare, a steep increase from Rs 12,000 per hectare earlier.
- Further substantial increase has been made for maintenance and inter-cropping interventions. A special assistance @ Rs 250 per plant is being given to replant old gardens for rejuvenation of old gardens.
- Seed gardens will be provided assistance up to Rs 100 lakhs for 15 hectares in the North-East and Andaman regions and Rs 80 lakhs for 15 hectares in the rest of India to address the issue of shortage of planting material in the country.

Q.91) Ans: B

Exp:

- Statement 1 is incorrect: National Legal Services Authority is a statutory body constituted under the Legal Services Authorities Act, 1987. Its aim is to ensure that opportunities for securing justice are not denied to any citizen by reasons of economic or other disabilities. It organises Lok Adalats for amicable settlement of disputes. As per section 3(2) of Legal Service Authorities Act,

the Chief Justice of India shall be the Patron-in-Chief. Second senior-most judge of the Supreme Court of India is the Executive-Chairman.

- Statement 2 is correct: The persons eligible for free legal service include:
 - o Women and children
 - o Members of SC/ST
 - o Industrial workmen
 - o Victims of mass disaster, violence, flood, drought, earthquake industrial disaster
 - o Disabled persons
 - o Persons in custody
 - o Persons whose annual income does not exceed ₹1 lakh (in the Supreme Court Legal Services Committee the limit is ₹1,25,000/-)
 - o Victims of trafficking in human beings or begar.

Q.92) Ans: A

Exp:

Option A is correct

About Ubharte Sitaare Fund:

- It is a Rs. 250 crore worth Alternative Investment Fund to provide debt and equity funding to export-oriented MSMEs as well as providing them with the required technological support to make their businesses more competitive.
- The fund has been set up jointly by Exim Bank and Small Industries Development Bank of India.
- The objective of the fund is
 - o To identify Indian enterprises with potential advantages by way of technology, products or processes along with export potential, but which are currently underperforming or unable to tap their latent potential to grow.
 - o The main purpose is to encourage MSMEs as they are vital to the economy in terms of creating jobs, fostering innovations and reviving the economy.
- The fund is a mix of structured support, both financial and advisory services.
- It will also have a Greenshoe Option of Rs 250 crore.

The Fund covers potential companies, across various sectors such as pharma, auto components, engineering solutions, agriculture, and software etc

Extra Edge by OnlyIAS

About Alternative Investment Fund:

- Alternative Investment Fund comprises pooled investment funds which invest in venture capital, private equity, hedge funds, managed futures etc.
- AIF does not include funds covered under the SEBI (Mutual Funds) Regulations, 1996, SEBI (Collective Investment Schemes) Regulations, 1999 or any other regulations of the Board to regulate fund management activities.
- Nonetheless, the alternative investment funds have to register with SEBI.

About Greenshoe Option:

A greenshoe option is an over-allotment option, which is a term that is commonly used to describe a special arrangement in a share offering for example an IPO (Initial Public Offering) that will enable the investment bank to support the share price after the offering without putting their own capital at risk.

Q.93) Ans: A

Exp:

- Statement 1 is correct: Atlantic Meridional Overturning Circulation (AMOC) is a large system of ocean currents that carry warm water from the tropics northwards into the North Atlantic. It is driven by differences in temperature and salt content – the water's density. As warm water flows northwards it cools and some evaporation occurs, which increases the amount of salt. Low temperature and a high salt content make the water denser, and this dense water sinks deep into the ocean. The cold, dense water slowly spreads southwards, several kilometres below the surface. Eventually, it gets pulled back to the surface and warms in a process called "upwelling" and the circulation is complete.
- Statement 2 is incorrect: If the AMOC collapsed, it would increase cooling of the Northern Hemisphere, sea level rise in the Atlantic, an overall fall in precipitation over Europe and North America and a shift in monsoons in South America and Africa.

Q.94) Ans: D

Exp:

Option D is correct (All the statements are incorrect)

About Great Hornbill:

- The great hornbill (*Buceros bicornis*) is also known as the concave-casqued hornbill, great Indian hornbill or great pied hornbill.
- It is found in the Indian subcontinent and Southeast Asia.
- The most prominent feature of the Great Hornbill is the bright yellow and black casque (hard structure on the head of some birds) on top of its massive bill.
- The Hornbill Festival celebrated in Nagaland is named after the bird – Hornbill, which is the most revered and admired bird for the Nagas.
- They are the cultural symbols of some ethnic communities in the northeast, specifically the Nyishi of Arunachal Pradesh.
- Great Hornbill is the state bird of Arunachal Pradesh and Kerala.
- It is listed in Schedule I of Wildlife Protection Act, 1972 and categorised as Vulnerable in IUCN Red list.
- Hornbills are hunted for their casques — upper beak — and feathers for adorning headgear. They are also poached for their meat and medicinal value of their body parts.

Q.95) Ans: D

Exp:

- Pair 1 is incorrect: Ningbo port is located in China and is the world's third-busiest container port. The Meishan terminal at Ningbo-Zhoushan port, which is south of Shanghai, accounts for over a fourth of the container cargo handled at the Chinese port. Meishan terminal mainly services trade destinations in North America and Europe.
- Pair 2 is incorrect: Cattle island is one of three islands in the Hirakud reservoir, Odisha. The island is a submerged hill, and before the construction of Hirakud Dam it was a developed village.
- Pair 3 is correct: Port-au-Prince is the capital of Haiti and was recently faced with an earthquake.



Q.96) Ans: D

Exp:

- Statement 1 is incorrect: E-Rupi is not a cryptocurrency. It is not even a digital form of Indian Rupee. E-RUPI is a prepaid, social-service voucher system launched by the Indian government. It is more of a digitized form of payment. It is a cashless and contactless method for digital payment. It is a Quick Response (QR) code or SMS string-based e-voucher, which is delivered to the mobile of the users. It means consumers can complete entire transactions on their cellphones, without having to transfer "money" in the traditional sense.
- Statement 2 is correct: e-RUPI is a digital payment solution developed with the support of DFS (Department of Financial Services) and NHA (National Health Authority) and is powered by National Payments Corporation of India (NPCI).
- Statement 3 is incorrect: RuPay Card is an Indian payment gateway. It is an Indian domestic card conceived and launched by NPCI in 2012. It was introduced in the Indian payment system to reduce the monopoly of foreign gateways like Visa and MasterCard. It also works like Visa or MasterCard and its commission is low. RuPay facilitates electronic payment at all Indian banks and financial institutions.

Q.97) Ans: A

Exp:

- Statement 1 is incorrect: Each member of the Security Council has one vote. Decisions of the Security Council on matters are made by an affirmative vote of nine members including the concurring votes of the permanent members. A "No" vote from one of the five permanent members blocks the passage of the resolution.
- Statement 2 is incorrect: Any member of the UN which is not a member of the Security Council may participate, without vote, in the discussion of any question brought before the Security Council whenever the latter considers that the interests of that member are specially affected.
- Statement 3 is correct: UN peacekeeping missions are mandated by the Security Council to provide security and political and peacebuilding support to countries in conflict or post-conflict

situations. They are guided by the principles of consent of the host country, impartiality, and non-use of force except in self-defence, defence of the mandate, or protection of civilians if so authorized by the Council.

Q.98) Ans: D

Exp:

- Statement 1 is incorrect: Fusion is among the most environmentally friendly sources of energy. Fusion reactions don't emit harmful toxins like carbon dioxide or other greenhouse gases into the atmosphere. Its major by-product is helium: an inert, non-toxic gas. Also, No long-lived radioactive waste is emitted.
- Statement 2 is incorrect: Fusion occurs when two atoms slam together to form a heavier atom, like when two hydrogen atoms fuse to form one helium atom. This is the same process that powers the sun and creates huge amounts of energy—several times greater than fission. Natural occurrence of the Fusion process occurs in stars, such as the sun. Whereas, Fission reaction does not normally occur in nature.

Q.99) Ans: C

Exp:

- Statement 1 is incorrect: Sunderbans is a vast contiguous mangrove forest ecosystem in the coastal region of Bay of Bengal spread over India and Bangladesh on the delta (world's largest) of the Ganges, Brahmaputra and Meghna rivers. It was designated as a Biosphere Reserve by UNESCO in 2001. Sundarban Wetland, India was recognised as the 'Wetland of International Importance' under the Ramsar Convention in January 2019.
- Statement 2 is incorrect: Kharai Camels/Swimming Camels are found only in Gujarat's Bhuj area. They are recognized as a separate breed of camels by the National Bureau of Animal Genetic Resources (NBAGR) for better conservation. This camel is adapted to the extreme climate of Rann of Kutch where shallow seas and high salinity is prevalent. The Kharai breed has the special ability to survive on both dry land and in the sea, making it an ecotonal breed.



- Statement 3 is correct: The Haliday Island Wildlife Sanctuary is a part of the Sundarbans Biosphere Reserve. It is one of the three Wildlife Sanctuaries of the Sundarbans Biosphere Reserve, the other two being the Sajnekhali Wildlife Sanctuary and the Lothian Wildlife Sanctuary. This Island is located on the river Matla, very close to the confluence of the Bay of Bengal.

Q.100) Ans: B

Exp:

- Statement 1 is correct: The Bengal School of Art commonly referred as Bengal School, was an art movement and a style of Indian painting that originated in Bengal, primarily Kolkata and Shantiniketan, and flourished throughout the Indian subcontinent, during the British Raj in the early 20th century. One of the prominent Paintings of Bengal mSchool is the "Bharat Mata" by Abanindranath Tagore. He made this painting during the Swadeshi Movement boycotting the British goods and creating a sense of patriotism and self-love for our country.
- Statement 2 is incorrect: Raja Ravi Verma Varma was patronised by Ayilyam Thirunal, the then ruler of Travancore. He was best known for uniting Hindu mythological subject matter with European realist. His paintings, which revolve mainly around the Puranas (ancient mythological stories) and the great Indian epics - Mahabharata and Ramayana
- Statement 3 is correct: The Japanese technique namely 'wash technique' was used in Bengal School. In painting it is a technique in which a paint brush that is very wet with solvent and holds a small load of paint or ink is applied to a wet or dry support such as paper or primed or raw canvas.